

leagues. The brachistochrone problem amounts to finding the curve along which a bead could slide over a frictionless wire in the least amount of time while under the influence only of gravity. The deadline for finding a solution was set at a year and a half. The problem was presented to Newton late one afternoon; by the next morning he had found the answer by inventing a new area of mathematics known as the calculus of variations. Although the solution was published anonymously at Newton's request, Bernoulli commented, "By the claw, the lion is revealed."

Concerning the successes of his own career, Newton wrote:

I do not know what I may appear to the world; but to myself I seem to have been only like a boy, playing on the seashore, and diverting myself, in now and then finding a smoother pebble or a prettier shell than ordinary, while the great ocean of truth lay all undiscovered before me.

Today, classical mechanics is described by Newton's three laws of motion, along with his universal *law of gravity*. Outside of the realms of atomic dimensions, velocities approaching the speed of light, or extreme gravitational forces, Newtonian physics has proved very successful in explaining the results of observations and experiments. Those regimes where Newtonian mechanics have been shown to be unsatisfactory will be discussed in later chapters.

Newton's first law of motion may be stated as follows:

Newton's First Law *The Law of Inertia.* An object at rest will remain at rest and an object in motion will remain in motion in a straight line at a constant speed unless acted upon by an external force.

To establish whether an object is actually moving, a reference frame must be established. In later chapters we will refer to reference frames that have the special property that the first law is valid; all such frames are known as **inertial reference frames**. Noninertial reference frames are accelerated with respect to inertial frames.

The first law may be restated in terms of the momentum of an object, $\mathbf{p} = m\mathbf{v}$, where m and $\mathbf{v}$ are mass and velocity, respectively.² Thus Newton's first law may be expressed as "*the momentum of an object remains constant unless it experiences an external force.*"³

The second law is actually a definition of the concept of force:

Newton's Second Law The *net* force (the sum of all forces) acting on an object is proportional to the object's mass and its resultant acceleration.

If an object is experiencing n forces, then the net force is given by

$$\mathbf{F}_{\text{net}} = \sum_{i=1}^n \mathbf{F}_i = m\mathbf{a}. \quad (2.9)$$

²Hereafter, all vectors will be indicated by boldface type. Vectors are quantities described by both a magnitude and a direction. Some texts use alternate notations for vectors, expressing them either as $\vec{v}$ or $\vec{\mathbf{v}}$.

³The law of inertia is an extension of the original concept developed by Galileo.

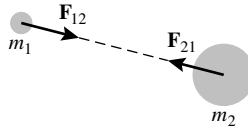


FIGURE 2.7 Newton's third law.

However, assuming that the mass is constant and using the definition $\mathbf{a} \equiv d\mathbf{v}/dt$, Newton's second law may also be expressed as

$$\mathbf{F}_{\text{net}} = m \frac{d\mathbf{v}}{dt} = \frac{d(m\mathbf{v})}{dt} = \frac{d\mathbf{p}}{dt}; \quad (2.10)$$

the net force on an object is equal to the time rate of change of its momentum, $\mathbf{p}$. $\mathbf{F}_{\text{net}} = d\mathbf{p}/dt$ actually represents the most general statement of the second law, allowing for a time variation in the mass of the object such as occurs with rocket propulsion.

The third law of motion is generally expressed as follows:

Newton's Third Law For every action there is an equal and opposite reaction.

In this law, *action* and *reaction* are to be interpreted as forces acting on different objects. Consider the force exerted *on* one object (object 1) *by* a second object (object 2), $\mathbf{F}_{12}$. Newton's third law states that the force exerted on object 2 by object 1, $\mathbf{F}_{21}$, must necessarily be of the same magnitude but in the opposite direction (see Fig. 2.7). Mathematically, the third law can be represented as

$$\mathbf{F}_{12} = -\mathbf{F}_{21}.$$

Newton's Law of Universal Gravitation

Using his three laws of motion along with Kepler's third law, Newton was able to find an expression describing the force that holds planets in their orbits. Consider the special case of *circular* orbital motion of a mass m about a much larger mass M ($M \gg m$). Allowing for a system of units other than years and astronomical units, Kepler's third law may be written as

$$P^2 = kr^3,$$

where r is the distance between the two objects and k is a constant of proportionality. Writing the period of the orbit in terms of the orbit's circumference and the constant velocity of m yields

$$P = \frac{2\pi r}{v},$$

and substituting into the prior equation gives

$$\frac{4\pi^2 r^2}{v^2} = kr^3.$$

Rearranging terms and multiplying both sides by m lead to the expression

$$m \frac{v^2}{r} = \frac{4\pi^2 m}{kr^2}.$$

The left-hand side of the equation may be recognized as the centripetal force for circular motion, so

$$F = \frac{4\pi^2 m}{kr^2}$$

must be the gravitational force keeping m in its orbit about M . However, Newton's third law states that the magnitude of the force exerted on M by m must equal the magnitude of the force exerted on m by M . Therefore, the form of the equation ought to be symmetric with respect to exchange of m and M , implying

$$F = \frac{4\pi^2 M}{k'r^2}.$$

Expressing this symmetry explicitly and grouping the remaining constants into a new constant, we have

$$F = \frac{4\pi^2 Mm}{k''r^2},$$

where $k = k''/M$ and $k' = k''/m$. Finally, introducing a new constant, $G \equiv 4\pi^2/k''$, we arrive at the form of the **Law of Universal Gravitation** found by Newton,

$$F = G \frac{Mm}{r^2}, \quad (2.11)$$

where $G = 6.673 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ (the **Universal Gravitational Constant**).⁴

Newton's law of gravity applies to any two objects having mass. In particular, for an extended object (as opposed to a point mass), the force exerted by that object on another extended object may be found by integrating over each of their mass distributions.

Example 2.2.1. The force exerted by a *spherically symmetric* object of mass M on a point mass m may be found by integrating over rings centered along a line connecting the point mass to the center of the extended object (see Fig. 2.8). In this way all points on a specific ring are located at the same distance from m . Furthermore, because of the symmetry of the ring, the gravitational force vector associated with it is oriented along the ring's central axis. Once a general description of the force due to one ring is determined, it is possible to add up the individual contributions from all such rings throughout the entire volume of the mass M . The result will be the force on m due to M .

continued

⁴At the time this text was written, the uncertainty in G was $\pm 0.010 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

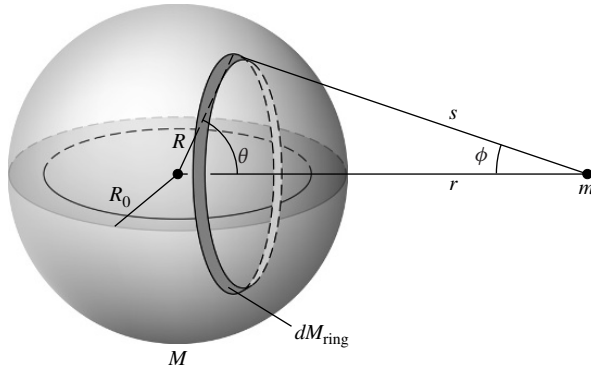


FIGURE 2.8 The gravitational effect of a spherically symmetric mass distribution.

Let r be the distance between the centers of the two masses, M and m . R_0 is the radius of the large mass, and s is the distance from the point mass to a point on the ring. Because of the symmetry of the problem, only the component of the gravitational force vector along the line connecting the centers of the two objects needs to be calculated; the perpendicular components will cancel. If dM_{ring} is the mass of the *ring* being considered, the force exerted by that ring on m is given by

$$dF_{\text{ring}} = G \frac{m dM_{\text{ring}}}{s^2} \cos \phi.$$

Assuming that the mass density, $\rho(R)$, of the extended object is a function of radius only and that the volume of the ring of thickness dR is dV_{ring} , we find that

$$\begin{aligned} dM_{\text{ring}} &= \rho(R) dV_{\text{ring}} \\ &= \rho(R) 2\pi R \sin \theta R d\theta dR \\ &= 2\pi R^2 \rho(R) \sin \theta dR d\theta. \end{aligned}$$

The cosine is given by

$$\cos \phi = \frac{r - R \cos \theta}{s},$$

where s may be found by the Pythagorean theorem:

$$s = \sqrt{(r - R \cos \theta)^2 + R^2 \sin^2 \theta} = \sqrt{r^2 - 2rR \cos \theta + R^2}.$$

Substituting into the expression for dF_{ring} , summing over all *rings* located at a distance R from the center of the mass M (i.e., integrating over all θ from 0 to π for constant R), and then summing over all resultant *shells* of radius R from $R = 0$ to $R = R_0$ give the total

force of gravity acting on the small mass m along the system's line of symmetry:

$$\begin{aligned} F &= Gm \int_0^{R_0} \int_0^\pi \frac{(r - R \cos \theta) \rho(R) 2\pi R^2 \sin \theta}{s^3} d\theta dR \\ &= 2\pi Gm \int_0^{R_0} \int_0^\pi \frac{r R^2 \rho(R) \sin \theta}{(r^2 + R^2 - 2rR \cos \theta)^{3/2}} d\theta dR \\ &\quad - 2\pi Gm \int_0^{R_0} \int_0^\pi \frac{R^3 \rho(R) \sin \theta \cos \theta}{(r^2 + R^2 - 2rR \cos \theta)^{3/2}} d\theta dR. \end{aligned}$$

The integrations over θ may be carried out by making the change of variable, $u \equiv s^2 = r^2 + R^2 - 2rR \cos \theta$. Then $\cos \theta = (r^2 + R^2 - u)/2rR$ and $\sin \theta d\theta = du/2rR$. After the appropriate substitutions and integration over the new variable u , the equation for the force becomes

$$F = \frac{Gm}{r^2} \int_0^{R_0} 4\pi R^2 \rho(R) dR.$$

Notice that the integrand is just the mass of a *shell* of thickness dR , having a volume dV_{shell} , or

$$dM_{\text{shell}} = 4\pi R^2 \rho(R) dR = \rho(R) dV_{\text{shell}}.$$

Therefore, the integrand gives the force on m due to a spherically symmetric mass shell of mass dM_{shell} as

$$dF_{\text{shell}} = \frac{Gm dM_{\text{shell}}}{r^2}.$$

The shell acts gravitationally as if its mass were located entirely at its center. Finally, integrating over the mass shells, we have that the force exerted on m by an extended, spherically symmetric mass distribution is directed along the line of symmetry between the two objects and is given by

$$F = G \frac{Mm}{r^2},$$

just the equation for the force of gravity between two point masses.

When an object is dropped near the surface of Earth, it accelerates toward the center of Earth at the rate $g = 9.80 \text{ m s}^{-2}$, the local acceleration of gravity. Using Newton's second law and his law of gravity, an expression for the acceleration of gravity may be found. If m is the mass of the falling object, $M_\oplus$ and $R_\oplus$ are the mass and radius of Earth, respectively, and h is the height of the object above Earth, then the force of gravity on m due to Earth is given by

$$F = G \frac{M_\oplus m}{(R_\oplus + h)^2}.$$

Assuming that m is near Earth's surface, then $h \ll R_\oplus$ and

$$F \simeq G \frac{M_\oplus m}{R_\oplus^2}.$$

However, $F = ma = mg$; thus

$$g = G \frac{M_\oplus}{R_\oplus^2}. \quad (2.12)$$

Substituting the values $M_\oplus = 5.9736 \times 10^{24}$ kg and $R_\oplus = 6.378136 \times 10^6$ m gives a value for g in agreement with the measured value.

The Orbit of the Moon

The famous story that an apple falling on Newton's head allowed him to immediately realize that gravity holds the Moon in its orbit is probably somewhat fanciful and inaccurate. However, he did demonstrate that, along with the acceleration of the falling apple, gravity was responsible for the motion of Earth's closest neighbor.

Example 2.2.2. Assuming for simplicity that the Moon's orbit is exactly circular, we can calculate the centripetal acceleration of the Moon rapidly. Recall that the centripetal acceleration of an object moving in a perfect circle is given by

$$a_c = \frac{v^2}{r}.$$

In this case, r is the distance from the center of Earth to the center of the Moon, $r = 3.84401 \times 10^8$ m, and v is the Moon's orbital velocity, given by

$$v = \frac{2\pi r}{P},$$

where $P = 27.3$ days $= 2.36 \times 10^6$ s is the sidereal orbital period of the Moon. Finding $v = 1.02$ km s⁻¹ gives a value for the centripetal acceleration of

$$a_c = 0.0027 \text{ m s}^{-2}.$$

The acceleration of the Moon caused by Earth's gravitational pull may also be calculated directly from

$$a_g = G \frac{M_\oplus}{r^2} = 0.0027 \text{ m s}^{-2},$$

in agreement with the value for the centripetal acceleration.

Work and Energy

In astrophysics, as in any area of physics, it is often very helpful to have some understanding of the energetics of specific physical phenomena in order to determine whether these processes are important in certain systems. Some models may be ruled out immediately if they are incapable of producing the amount of energy observed. Energy arguments also often result in simpler solutions to particular problems. For example, in the evolution of a planetary atmosphere, the possibility of a particular component of the atmosphere escaping must be considered. Such a consideration is based on a calculation of the escape speed of the gas particles.

The amount of energy (the work) necessary to raise an object of mass m a height h against a gravitational force is equal to the change in the *potential energy* of the system. Generally, the change in potential energy resulting from a change in position between two points is given by

$$U_f - U_i = \Delta U = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r}, \quad (2.13)$$

where $\mathbf{F}$ is the force vector, $\mathbf{r}_i$ and $\mathbf{r}_f$ are the initial and final position vectors, respectively, and $d\mathbf{r}$ is the infinitesimal change in the position vector for some general coordinate system (see Fig. 2.9). If the gravitational force on m is due to a mass M located at the origin, then $\mathbf{F}$ is directed inward toward M , $d\mathbf{r}$ is directed outward, $\mathbf{F} \cdot d\mathbf{r} = -F dr$, and the change in potential energy becomes

$$\Delta U = \int_{r_i}^{r_f} G \frac{Mm}{r^2} dr.$$

Evaluating the integral, we have

$$U_f - U_i = -GMm \left(\frac{1}{r_f} - \frac{1}{r_i} \right).$$

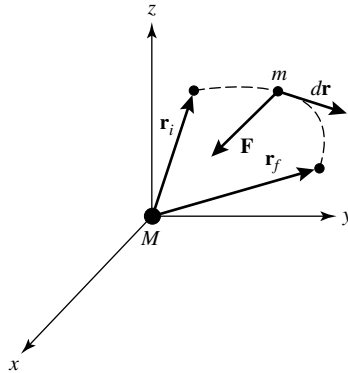


FIGURE 2.9 Gravitational potential energy. The amount of work done depends on the direction of motion relative to the direction of the force vector.

Since only *relative changes* in potential energy are physically meaningful, a reference position where the potential energy is defined as being identically zero may be chosen. If, for a specific gravitational system, it is assumed that the potential energy goes to zero at infinity, letting r_f approach infinity ($r_f \rightarrow \infty$) and dropping the subscripts for simplicity give

$$\boxed{U = -G \frac{Mm}{r}}. \quad (2.14)$$

Of course, the process can be reversed: The force may be found by *differentiating* the gravitational potential energy. For forces that depend only on r ,

$$F = -\frac{\partial U}{\partial r}. \quad (2.15)$$

In a general three-dimensional description, $\mathbf{F} = -\nabla U$, where ∇U represents the *gradient* of U . In rectangular coordinates this becomes

$$\mathbf{F} = -\frac{\partial U}{\partial x} \hat{\mathbf{i}} - \frac{\partial U}{\partial y} \hat{\mathbf{j}} - \frac{\partial U}{\partial z} \hat{\mathbf{k}}.$$

Work must be performed on a massive object if its *speed*, $|\mathbf{v}|$, is to be changed. This can be seen by rewriting the work integral, first in terms of time, then speed:

$$\begin{aligned} W &\equiv -\Delta U \\ &= \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{t_i}^{t_f} \frac{d\mathbf{p}}{dt} \cdot (\mathbf{v} dt) \\ &= \int_{t_i}^{t_f} m \frac{d\mathbf{v}}{dt} \cdot (\mathbf{v} dt) \\ &= \int_{t_i}^{t_f} m \left(\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} \right) dt \\ &= \int_{t_i}^{t_f} m \frac{d(\frac{1}{2}v^2)}{dt} dt \\ &= \int_{v_i}^{v_f} m d\left(\frac{1}{2}v^2\right) \\ &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2. \end{aligned}$$

We may now identify the quantity

$$K = \frac{1}{2}mv^2 \quad (2.16)$$

as the **kinetic energy** of the object. Thus work done on the particle results in an equivalent change in the particle's kinetic energy. This statement is simply one example of the **conservation of energy**, a concept that is encountered frequently in all areas of physics.

Consider a particle of mass m that has an initial velocity $\mathbf{v}$ and is at a distance r from the center of a larger mass M , such as Earth. How fast must the mass be moving upward to completely escape the pull of gravity? To calculate the **escape speed**, energy conservation may be used directly. The total initial mechanical energy of the particle (both kinetic and potential) is given by

$$E = \frac{1}{2}mv^2 - G\frac{Mm}{r}.$$

Assume that, in the critical case, the final velocity of the mass will be zero at a position infinitely far from M , implying that both the kinetic and potential energy will become zero. Clearly, by conservation of energy, the *total* energy of the particle must be identically zero at all times. Thus

$$\frac{1}{2}mv^2 = G\frac{Mm}{r},$$

which may be solved immediately for the initial speed of m to give

$$v_{\text{esc}} = \sqrt{2GM/r}. \quad (2.17)$$

Notice that the mass of the escaping object does not enter into the final expression for the escape speed. Near the surface of Earth, $v_{\text{esc}} = 11.2 \text{ km s}^{-1}$.

2.3 ■ KEPLER'S LAWS DERIVED

Although Kepler did finally determine that the geometry of planetary motion was in the more general form of an ellipse rather than circular motion, he was unable to explain the nature of the force that kept the planets moving in their precise patterns. Not only was Newton successful in quantifying that force, he was also able to generalize Kepler's work, deriving the empirical laws of planetary motion from the gravitational force law. The derivation of Kepler's laws represented a crucial step in the development of modern astrophysics.

The Center-of-Mass Reference Frame

However, before proceeding onward to derive Kepler's laws, it will be useful to examine more closely the dynamics of orbital motion. An interacting two-body problem, such as

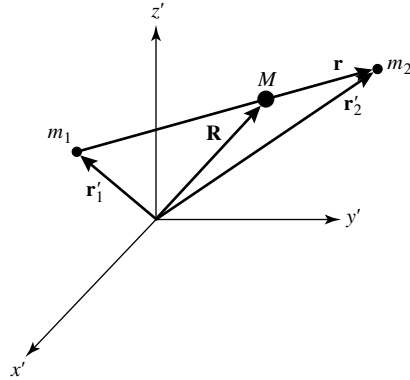


FIGURE 2.10 A general Cartesian coordinate system indicating the positions of m_1 , m_2 , and the center of mass (located at M).

binary orbits, or the more general many-body problem (often called the N -body problem), is most easily done in the reference frame of the **center of mass**.

Figure 2.10 shows two objects of masses m_1 and m_2 at positions $\mathbf{r}'_1$ and $\mathbf{r}'_2$, respectively, with the displacement vector from $\mathbf{r}'_1$ to $\mathbf{r}'_2$ given by

$$\mathbf{r} = \mathbf{r}'_2 - \mathbf{r}'_1.$$

Define a position vector $\mathbf{R}$ to be a *weighted* average of the position vectors of the individual masses,

$$\mathbf{R} \equiv \frac{m_1 \mathbf{r}'_1 + m_2 \mathbf{r}'_2}{m_1 + m_2}. \quad (2.18)$$

Of course, this definition can be immediately generalized to the case of n objects,

$$\mathbf{R} \equiv \frac{\sum_{i=1}^n m_i \mathbf{r}'_i}{\sum_{i=1}^n m_i}.$$

Rewriting the equation, we have

$$\sum_{i=1}^n m_i \mathbf{R} = \sum_{i=1}^n m_i \mathbf{r}'_i.$$

Then, if we define M to be the total mass of the system, $M \equiv \sum_{i=1}^n m_i$, the previous equation becomes

$$M \mathbf{R} = \sum_{i=1}^n m_i \mathbf{r}'_i.$$

Assuming that the individual masses do not change, differentiating both sides with respect

to time gives

$$M \frac{d\mathbf{R}}{dt} = \sum_{i=1}^n m_i \frac{d\mathbf{r}'_i}{dt}$$

or

$$M\mathbf{V} = \sum_{i=1}^n m_i \mathbf{v}'_i.$$

The right-hand side is the sum of the linear momenta of every particle in the system, so the total linear momentum of the system may be treated as though *all* of the mass were located at $\mathbf{R}$, moving with a velocity $\mathbf{V}$. Thus $\mathbf{R}$ is the position of the center of mass of the system, and $\mathbf{V}$ is the center-of-mass velocity. Letting $\mathbf{P} \equiv M\mathbf{V}$ be the linear momentum of the center of mass and $\mathbf{p}'_i \equiv m_i \mathbf{v}'_i$ be the linear momentum of an individual particle i , and again differentiating both sides with respect to time, yields

$$\frac{d\mathbf{P}}{dt} = \sum_{i=1}^n \frac{d\mathbf{p}'_i}{dt}.$$

If we assume that *all* of the forces acting on individual particles in the system are due to other particles contained within the system, Newton's third law requires that the total force must be zero. This constraint exists because of the equal magnitudes and opposite directions of action–reaction pairs. Of course, the momentum of individual masses may change. Using center-of-mass quantities, we find that the total (or net) force on the system is

$$\mathbf{F} = \frac{d\mathbf{P}}{dt} = M \frac{d^2\mathbf{R}}{dt^2} = 0.$$

Therefore, the center of mass will not accelerate if no external forces exist. This implies that a reference frame associated with the center of mass must be an inertial reference frame and that the N -body problem may be simplified by choosing a coordinate system for which the center of mass is at rest at $\mathbf{R} = 0$.

If we choose a center-of-mass reference frame for a binary system, depicted in Fig. 2.11 ($\mathbf{R} = 0$), Eq. (2.18) becomes

$$\frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2} = 0, \quad (2.19)$$

where the primes have been dropped, indicating center-of-mass coordinates. Both $\mathbf{r}_1$ and $\mathbf{r}_2$ may now be rewritten in terms of the displacement vector, $\mathbf{r}$. Substituting $\mathbf{r}_2 = \mathbf{r}_1 + \mathbf{r}$ gives

$$\mathbf{r}_1 = -\frac{m_2}{m_1 + m_2} \mathbf{r} \quad (2.20)$$

$$\mathbf{r}_2 = \frac{m_1}{m_1 + m_2} \mathbf{r}. \quad (2.21)$$

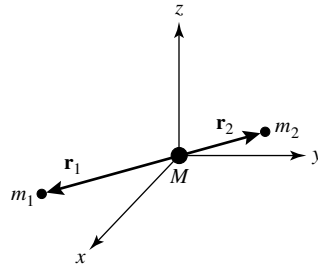


FIGURE 2.11 The center-of-mass reference frame for a binary orbit, with the center of mass fixed at the origin of the coordinate system.

Next, define the **reduced mass** to be

$$\mu \equiv \frac{m_1 m_2}{m_1 + m_2}. \quad (2.22)$$

Then $\mathbf{r}_1$ and $\mathbf{r}_2$ become

$$\mathbf{r}_1 = -\frac{\mu}{m_1} \mathbf{r} \quad (2.23)$$

$$\mathbf{r}_2 = \frac{\mu}{m_2} \mathbf{r}. \quad (2.24)$$

The convenience of the center-of-mass reference frame becomes evident when the total energy and orbital angular momentum of the system are considered. Including the necessary kinetic energy and gravitational potential energy terms, the total energy may be expressed as

$$E = \frac{1}{2} m_1 |\mathbf{v}_1|^2 + \frac{1}{2} m_2 |\mathbf{v}_2|^2 - G \frac{m_1 m_2}{|\mathbf{r}_2 - \mathbf{r}_1|}.$$

Substituting the relations for $\mathbf{r}_1$ and $\mathbf{r}_2$, along with the expression for the total mass of the system and the definition for the reduced mass, gives

$$E = \frac{1}{2} \mu v^2 - G \frac{M \mu}{r}, \quad (2.25)$$

where $v = |\mathbf{v}|$ and $\mathbf{v} \equiv d\mathbf{r}/dt$. We have also used the notation $r = |\mathbf{r}_2 - \mathbf{r}_1|$. The total energy of the system is equal to the kinetic energy of the reduced mass, plus the potential energy of the reduced mass moving about a mass M , assumed to be located and fixed at the origin. The distance between μ and M is equal to the separation between the objects of masses m_1 and m_2 .

Similarly, the total orbital angular momentum,

$$\mathbf{L} = m_1 \mathbf{r}_1 \times \mathbf{v}_1 + m_2 \mathbf{r}_2 \times \mathbf{v}_2$$

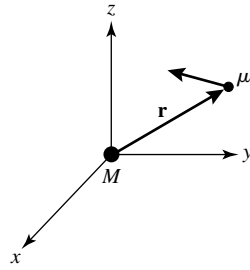


FIGURE 2.12 A binary orbit may be reduced to the equivalent problem of calculating the motion of the reduced mass, μ , about the total mass, M , located at the origin.

becomes

$$\mathbf{L} = \mu \mathbf{r} \times \mathbf{v} = \mathbf{r} \times \mathbf{p}, \quad (2.26)$$

where $\mathbf{p} \equiv \mu \mathbf{v}$. The total orbital angular momentum equals the angular momentum of the reduced mass only. *In general, the two-body problem may be treated as an equivalent one-body problem with the reduced mass μ moving about a fixed mass M at a distance r (see Fig. 2.12).*

The Derivation of Kepler's First Law

To obtain Kepler's laws, we begin by considering the effect of gravitation on the orbital angular momentum of a planet. Using center-of-mass coordinates and evaluating the time derivative of the orbital angular momentum of the reduced mass (Eq. 2.26) give

$$\frac{d\mathbf{L}}{dt} = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} = \mathbf{v} \times \mathbf{p} + \mathbf{r} \times \mathbf{F},$$

the second expression arising from the definition of velocity and Newton's second law. Notice that because $\mathbf{v}$ and $\mathbf{p}$ are in the same direction, their cross product is identically zero. Similarly, since $\mathbf{F}$ is a central force directed inward along $\mathbf{r}$, the cross product of $\mathbf{r}$ and $\mathbf{F}$ is also zero. The result is an important general statement concerning angular momentum:

$$\frac{d\mathbf{L}}{dt} = 0, \quad (2.27)$$

the angular momentum of a system is a constant for a central force law. Equation (2.26) further shows that the position vector $\mathbf{r}$ is always perpendicular to the constant angular momentum vector $\mathbf{L}$, meaning that the orbit of the reduced mass lies in a plane perpendicular to $\mathbf{L}$.

Using the radial unit vector $\hat{\mathbf{r}}$ (so $\mathbf{r} = r\hat{\mathbf{r}}$), we can write the angular momentum vector in an alternative form as

$$\begin{aligned} \mathbf{L} &= \mu \mathbf{r} \times \mathbf{v} \\ &= \mu r \hat{\mathbf{r}} \times \frac{d}{dt} (r \hat{\mathbf{r}}) \end{aligned}$$

$$\begin{aligned}
&= \mu r \hat{\mathbf{r}} \times \left(\frac{dr}{dt} \hat{\mathbf{r}} + r \frac{d}{dt} \hat{\mathbf{r}} \right) \\
&= \mu r^2 \hat{\mathbf{r}} \times \frac{d}{dt} \hat{\mathbf{r}}.
\end{aligned}$$

(The last result comes from the fact that $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = 0$.) In vector form, the acceleration of the reduced mass due to the gravitational force exerted by M is

$$\mathbf{a} = -\frac{GM}{r^2} \hat{\mathbf{r}}.$$

Taking the vector cross product of the acceleration of the reduced mass with its own orbital angular momentum gives

$$\mathbf{a} \times \mathbf{L} = -\frac{GM}{r^2} \hat{\mathbf{r}} \times \left(\mu r^2 \hat{\mathbf{r}} \times \frac{d}{dt} \hat{\mathbf{r}} \right) = -GM\mu \hat{\mathbf{r}} \times \left(\hat{\mathbf{r}} \times \frac{d}{dt} \hat{\mathbf{r}} \right).$$

Applying the vector identity $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$ results in

$$\mathbf{a} \times \mathbf{L} = -GM\mu \left[\left(\hat{\mathbf{r}} \cdot \frac{d}{dt} \hat{\mathbf{r}} \right) \hat{\mathbf{r}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) \frac{d}{dt} \hat{\mathbf{r}} \right].$$

Because $\hat{\mathbf{r}}$ is a unit vector, $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = 1$ and

$$\frac{d}{dt}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) = 2 \hat{\mathbf{r}} \cdot \frac{d}{dt} \hat{\mathbf{r}} = 0.$$

As a result,

$$\mathbf{a} \times \mathbf{L} = GM\mu \frac{d}{dt} \hat{\mathbf{r}}$$

or, by referring to Eq. (2.27),

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = \frac{d}{dt}(GM\mu \hat{\mathbf{r}}).$$

Integrating with respect to time then yields

$$\mathbf{v} \times \mathbf{L} = GM\mu \hat{\mathbf{r}} + \mathbf{D}, \quad (2.28)$$

where $\mathbf{D}$ is a constant vector. Because $\mathbf{v} \times \mathbf{L}$ and $\hat{\mathbf{r}}$ both lie in the orbital plane, so must $\mathbf{D}$. Furthermore, the magnitude of the left-hand side will be greatest at perihelion when the velocity of the reduced mass is a maximum. Moreover, the magnitude of the right-hand side is greatest when $\hat{\mathbf{r}}$ and $\mathbf{D}$ point in the same direction. Therefore, $\mathbf{D}$ is directed toward perihelion. As shown below, the magnitude of $\mathbf{D}$ determines the eccentricity of the orbit.

We next take the vector dot product of Eq. (2.28) with the position vector $\mathbf{r} = r \hat{\mathbf{r}}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{L}) = GM\mu r \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} + \mathbf{r} \cdot \mathbf{D}.$$

Invoking the vector identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$ gives

$$(\mathbf{r} \times \mathbf{v}) \cdot \mathbf{L} = GM\mu r + rD \cos \theta.$$

Finally, recalling the definition of angular momentum (Eq. 2.26), we obtain

$$\frac{L^2}{\mu} = GM\mu r \left(1 + \frac{D \cos \theta}{GM\mu} \right),$$

where θ is the angle of the reduced mass as measured from the direction to perihelion. Defining $e \equiv D/GM\mu$ and solving for r , we find

Kepler's First Law (revisited)

$$r = \frac{L^2/\mu^2}{GM(1 + e \cos \theta)}. \quad (2.29)$$

This is exactly the equation of a conic section, as may be seen by comparing Eq. (2.29) with Eqs. (2.3), (2.7), and (2.8) for an ellipse, parabola, and hyperbola, respectively. *The path of the reduced mass about the center of mass under the influence of gravity (or any other inverse-square force) is a conic section.* Elliptical orbits result from an attractive r^{-2} central-force law such as gravity when the total energy of the system is less than zero (a bound system), parabolic trajectories are obtained when the energy is identically zero, and hyperbolic paths result from an unbounded system with an energy that is greater than zero.

When Eq. (2.29) is translated back to a physical reference frame on the sky, we find that Kepler's first law for bound planetary orbits may be stated as: *Both objects in a binary orbit move about the center of mass in ellipses, with the center of mass occupying one focus of each ellipse.* Newton was able to demonstrate the elliptical behavior of planetary motion and found that Kepler's first law must be generalized somewhat: The center of mass of the system, rather than the exact center of the Sun, is actually located at the focus of the ellipse. For our Solar System, such a mistake is understandable, since the largest of the planets, Jupiter, has only 1/1000 the mass of the Sun. This places the center of mass of the Sun–Jupiter system near the surface of the Sun. Having used the naked-eye data of Tycho, Kepler can be forgiven for not realizing his error.

For the case of closed planetary orbits, comparing Eqs. (2.3) and (2.29) shows that the total orbital angular momentum of the system is

$$L = \mu \sqrt{GMa(1 - e^2)}. \quad (2.30)$$

Note that L is a maximum for purely circular motion ($e = 0$) and goes to zero as the eccentricity approaches unity, as expected.

The Derivation of Kepler's Second Law

To derive Kepler's second law, which relates the area of a section of an ellipse to a time interval, we begin by considering the infinitesimal area element in polar coordinates, as

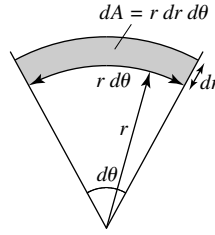


FIGURE 2.13 The infinitesimal area element in polar coordinates.

shown in Fig. 2.13:

$$dA = dr (r d\theta) = r dr d\theta.$$

If we integrate from the principal focus of the ellipse to a specific distance, r , the area swept out by an infinitesimal change in θ becomes

$$dA = \frac{1}{2} r^2 d\theta.$$

Therefore, the time rate of change in area swept out by a line joining a point on the ellipse to the focus becomes

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}. \quad (2.31)$$

Now the orbital velocity, $\mathbf{v}$, may be expressed in two components, one directed along $\mathbf{r}$ and the other perpendicular to $\mathbf{r}$. Letting $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ be the unit vectors along $\mathbf{r}$ and its normal, respectively, $\mathbf{v}$ may be written as (see Fig. 2.14)

$$\mathbf{v} = \mathbf{v}_r + \mathbf{v}_\theta = \frac{dr}{dt} \hat{\mathbf{r}} + r \frac{d\theta}{dt} \hat{\boldsymbol{\theta}}.$$

Substituting v_θ into Eq. (2.31) gives

$$\frac{dA}{dt} = \frac{1}{2} r v_\theta.$$

Since $\mathbf{r}$ and $\mathbf{v}_\theta$ are perpendicular,

$$r v_\theta = |\mathbf{r} \times \mathbf{v}| = \left| \frac{\mathbf{L}}{\mu} \right| = \frac{L}{\mu}.$$

Finally, the time derivative of the area becomes

Kepler's Second Law (revisited)

$$\boxed{\frac{dA}{dt} = \frac{1}{2} \frac{L}{\mu}}. \quad (2.32)$$

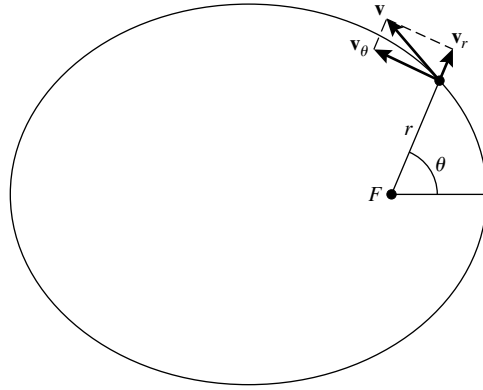


FIGURE 2.14 The velocity vector for elliptical motion in polar coordinates.

It has already been shown that the orbital angular momentum is a constant, so *the time rate of change of the area swept out by a line connecting a planet to the focus of an ellipse is a constant*, one-half of the orbital angular momentum per unit mass. This is just Kepler's second law.

Expressions for the speed of the reduced mass at perihelion ($\theta = 0$) and aphelion ($\theta = \pi$) may be easily obtained from Eq. (2.29). Since at both perihelion and aphelion, $\mathbf{r}$ and $\mathbf{v}$ are perpendicular, the magnitude of the angular momentum at these points simply becomes

$$L = \mu r v.$$

Eq. (2.29) at perihelion may thus be written as

$$r_p = \frac{(\mu r_p v_p)^2 / \mu^2}{GM(1 + e)},$$

whereas at aphelion

$$r_a = \frac{(\mu r_a v_a)^2 / \mu^2}{GM(1 - e)}.$$

Recalling from Example 2.1.1 that $r_p = a(1 - e)$ at perihelion and $r_a = a(1 + e)$ at aphelion, we immediately obtain

$$v_p^2 = \frac{GM(1 + e)}{r_p} = \frac{GM}{a} \left(\frac{1 + e}{1 - e} \right) \quad (2.33)$$

at perihelion and

$$v_a^2 = \frac{GM(1 - e)}{r_a} = \frac{GM}{a} \left(\frac{1 - e}{1 + e} \right). \quad (2.34)$$

at aphelion.

The total orbital energy may be found as well:

$$E = \frac{1}{2}\mu v_p^2 - G \frac{M\mu}{r_p}.$$

Making the appropriate substitutions, and after some rearrangement,

$$E = -G \frac{M\mu}{2a} = -G \frac{m_1 m_2}{2a}. \quad (2.35)$$

The total energy of a binary orbit depends only on the semimajor axis a and is exactly one-half the time-averaged potential energy of the system,

$$E = \frac{1}{2} \langle U \rangle,$$

where $\langle U \rangle$ denotes an average over one orbital period.⁵ This is one example of the *virial theorem*, a general property of gravitationally bound systems. The virial theorem will be discussed in detail in Section 2.4.

A useful expression for the velocity of the reduced mass (or the relative velocity of m_1 and m_2) may be found directly by using the conservation of energy and equating the total orbital energy to the sum of the kinetic and potential energies:

$$-G \frac{M\mu}{2a} = \frac{1}{2}\mu v^2 - G \frac{M\mu}{r}.$$

Using the identity $M = m_1 + m_2$, this simplifies to give

$$v^2 = G (m_1 + m_2) \left(\frac{2}{r} - \frac{1}{a} \right). \quad (2.36)$$

This expression could also have been obtained directly by adding the vector components of orbital velocity. Calculating $\mathbf{v}_r$, $\mathbf{v}_\theta$, and v^2 will be left as exercises.

The Derivation of Kepler's Third Law

We are finally in a position to derive the last of Kepler's laws. Integrating the mathematical expression for Kepler's second law (Eq. 2.32) over one orbital period, P , gives the result

$$A = \frac{1}{2} \frac{L}{\mu} P.$$

Here the mass m orbiting about a much larger *fixed* mass M has been replaced by the more general reduced mass μ orbiting about the center of mass. Substituting the area of an ellipse,

⁵The proof that $\langle U \rangle = -GM\mu/a$ is left as an exercise. Note that the time average, $\langle 1/r \rangle$, is equal to $1/a$, but $\langle r \rangle \neq a$.

$A = \pi ab$, squaring the equation, and rearranging, we obtain the expression

$$P^2 = \frac{4\pi^2 a^2 b^2 \mu^2}{L^2}.$$

Finally, using Eq. (2.2) and the expression for the total orbital angular momentum (Eq. 2.30), the last equation simplifies to become

Kepler's Third Law (revisited)

$$P^2 = \frac{4\pi^2}{G(m_1 + m_2)} a^3. \quad (2.37)$$

This is the general form of Kepler's third law. Not only did Newton demonstrate the relationship between the semimajor axis of an elliptical orbit and the orbital period, he also found a term not discovered empirically by Kepler, the square of the orbital period is inversely proportional to the total mass of the system. Once again Kepler can be forgiven for not noticing the effect. Tycho's data were for our Solar System only, and because the Sun's mass $M_\odot$ is so much greater than the mass of any of the planets, $M_\odot + m_{\text{planet}} \simeq M_\odot$. Expressing P in years and a in astronomical units gives a value of unity for the collection of constants (including the Sun's mass).⁶

The importance to astronomy of Newton's form of Kepler's third law cannot be overstated. This law provides the most direct way of obtaining masses of celestial objects, a critical parameter in understanding a wide range of phenomena. Kepler's laws, as derived by Newton, apply equally well to planets orbiting the Sun, moons orbiting planets, stars in orbit about one another, and galaxy–galaxy orbits. Knowledge of the period of an orbit and the semimajor axis of the ellipse yields the total mass of the system. If relative distances to the center of mass are also known, the individual masses may be determined using Eq. (2.19).

Example 2.3.1. The orbital sidereal period of Io, one of the four Galilean moons of Jupiter, is 1.77 days $= 1.53 \times 10^5$ s and the semimajor axis of its orbit is 4.22×10^8 m. Assuming that the mass of Io is insignificant compared to that of Jupiter, the mass of the planet may be estimated using Kepler's third law:

$$M_{\text{Jupiter}} = \frac{4\pi^2}{G} \frac{a^3}{P^2} = 1.90 \times 10^{27} \text{ kg} = 318 M_\oplus.$$

Appendix J describes a simple computer program that makes use of many of the ideas discussed in this chapter. *Orbit* will calculate, as a function of time, the location of a small mass that is orbiting about a much larger star (or it may be thought of as calculating the motion of the reduced mass about the total mass). Data generated by *Orbit* were used to produce Fig. 2.2.

⁶In 1621 Kepler was able to demonstrate that the four Galilean moons also obeyed his third law in the form $P^2 = ka^3$, where the constant k differed from unity. He did not attribute the fact that $k \neq 1$ to mass, however.

2.4 ■ THE VIRIAL THEOREM

In the last section we found that the total energy of the binary orbit was just one-half of the time-averaged gravitational potential energy (Eq. 2.35), or $E = \langle U \rangle / 2$. Since the total energy of the system is negative, the system is necessarily bound. For gravitationally bound systems in equilibrium, it can be shown that the total energy is always one-half of the time-averaged potential energy; this is known as the **virial theorem**.

To prove the virial theorem, begin by considering the quantity

$$Q \equiv \sum_i \mathbf{p}_i \cdot \mathbf{r}_i,$$

where $\mathbf{p}_i$ and $\mathbf{r}_i$ are the linear momentum and position vectors for particle i in some inertial reference frame, and the sum is taken to be over all particles in the system. The time derivative of Q is

$$\frac{dQ}{dt} = \sum_i \left(\frac{d\mathbf{p}_i}{dt} \cdot \mathbf{r}_i + \mathbf{p}_i \cdot \frac{d\mathbf{r}_i}{dt} \right). \quad (2.38)$$

Now, the left-hand side of the expression is just

$$\frac{dQ}{dt} = \frac{d}{dt} \sum_i m_i \frac{d\mathbf{r}_i}{dt} \cdot \mathbf{r}_i = \frac{d}{dt} \sum_i \frac{1}{2} \frac{d}{dt} (m_i r_i^2) = \frac{1}{2} \frac{d^2 I}{dt^2},$$

where

$$I = \sum_i m_i r_i^2$$

is the *moment of inertia* of the collection of particles. Substituting back into Eq. (2.38),

$$\frac{1}{2} \frac{d^2 I}{dt^2} - \sum_i \mathbf{p}_i \cdot \frac{d\mathbf{r}_i}{dt} = \sum_i \frac{d\mathbf{p}_i}{dt} \cdot \mathbf{r}_i. \quad (2.39)$$

The second term on the left-hand side is just

$$- \sum_i \mathbf{p}_i \cdot \frac{d\mathbf{r}_i}{dt} = - \sum_i m_i \mathbf{v}_i \cdot \mathbf{v}_i = -2 \sum_i \frac{1}{2} m_i v_i^2 = -2K,$$

twice the negative of the total kinetic energy of the system. If we use Newton's second law, Eq. (2.39) becomes

$$\frac{1}{2} \frac{d^2 I}{dt^2} - 2K = \sum_i \mathbf{F}_i \cdot \mathbf{r}_i. \quad (2.40)$$

The right-hand side of this expression is known as the *virial of Clausius*, named after the physicist who first found this important energy relation.

If $\mathbf{F}_{ij}$ represents the force of interaction between two particles in the system (actually the force on i due to j), then, considering all of the possible forces acting on i ,

$$\sum_i \mathbf{F}_i \cdot \mathbf{r}_i = \sum_i \left(\sum_{\substack{j \\ j \neq i}} \mathbf{F}_{ij} \right) \cdot \mathbf{r}_i.$$

Rewriting the position vector of particle i as $\mathbf{r}_i = \frac{1}{2}(\mathbf{r}_i + \mathbf{r}_j) + \frac{1}{2}(\mathbf{r}_i - \mathbf{r}_j)$, we find

$$\sum_i \mathbf{F}_i \cdot \mathbf{r}_i = \frac{1}{2} \sum_i \left(\sum_{\substack{j \\ j \neq i}} \mathbf{F}_{ij} \right) \cdot (\mathbf{r}_i + \mathbf{r}_j) + \frac{1}{2} \sum_i \left(\sum_{\substack{j \\ j \neq i}} \mathbf{F}_{ij} \right) \cdot (\mathbf{r}_i - \mathbf{r}_j).$$

From Newton's third law, $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$, implying that the first term on the right-hand side is zero, by symmetry. Thus the virial of Clausius may be expressed as

$$\sum_i \mathbf{F}_i \cdot \mathbf{r}_i = \frac{1}{2} \sum_i \sum_{\substack{j \\ j \neq i}} \mathbf{F}_{ij} \cdot (\mathbf{r}_i - \mathbf{r}_j). \quad (2.41)$$

If it is assumed that the only contribution to the force is the result of the gravitational interaction between massive particles included in the system, then $\mathbf{F}_{ij}$ is

$$\mathbf{F}_{ij} = G \frac{m_i m_j}{r_{ij}^2} \hat{\mathbf{r}}_{ij},$$

where $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ is the separation between particles i and j , and $\hat{\mathbf{r}}_{ij}$ is the unit vector directed from i to j :

$$\hat{\mathbf{r}}_{ij} \equiv \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}}.$$

Substituting the gravitational force into Eq. (2.41) gives

$$\begin{aligned} \sum_i \mathbf{F}_i \cdot \mathbf{r}_i &= -\frac{1}{2} \sum_i \sum_{\substack{j \\ j \neq i}} G \frac{m_i m_j}{r_{ij}^3} (\mathbf{r}_j - \mathbf{r}_i)^2 \\ &= -\frac{1}{2} \sum_i \sum_{\substack{j \\ j \neq i}} G \frac{m_i m_j}{r_{ij}}. \end{aligned} \quad (2.42)$$

The quantity

$$-G \frac{m_i m_j}{r_{ij}}$$

is just the potential energy U_{ij} between particles i and j . Note, however, that

$$-G \frac{m_j m_i}{r_{ji}}$$

also represents the same potential energy term and is included in the double sum as well, so the right-hand side of Eq. (2.42) includes the potential interaction between each pair of particles twice. Considering the factor of 1/2, Eq. (2.42) simply becomes

$$\sum_i \mathbf{F}_i \cdot \mathbf{r}_i = -\frac{1}{2} \sum_i \sum_{\substack{j \\ j \neq i}} G \frac{m_i m_j}{r_{ij}} = \frac{1}{2} \sum_i \sum_{\substack{j \\ j \neq i}} U_{ij} = U, \quad (2.43)$$

the total potential energy of the system of particles. Finally, substituting into Eq. (2.40) and taking the average with respect to time give

$$\frac{1}{2} \left\langle \frac{d^2 I}{dt^2} \right\rangle - 2 \langle K \rangle = \langle U \rangle. \quad (2.44)$$

The average of $d^2 I / dt^2$ over some time interval τ is just

$$\begin{aligned} \left\langle \frac{d^2 I}{dt^2} \right\rangle &= \frac{1}{\tau} \int_0^\tau \frac{d^2 I}{dt^2} dt \\ &= \frac{1}{\tau} \left(\left. \frac{dI}{dt} \right|_\tau - \left. \frac{dI}{dt} \right|_0 \right). \end{aligned} \quad (2.45)$$

If the system is periodic, as in the case for orbital motion, then

$$\left. \frac{dI}{dt} \right|_\tau = \left. \frac{dI}{dt} \right|_0$$

and the average over one period will be zero. Even if the system being considered is not strictly periodic, the average will still approach zero when evaluated over a sufficiently long period of time (i.e., $\tau \rightarrow \infty$), assuming of course that dI/dt is bounded. This would describe, for example, a system that has reached an equilibrium or steady-state configuration. In either case, we now have $\langle d^2 I / dt^2 \rangle = 0$, so

$$\boxed{-2 \langle K \rangle = \langle U \rangle.} \quad (2.46)$$

This result is one form of the virial theorem. The theorem may also be expressed in terms of the total energy of the system by using the relation $\langle E \rangle = \langle K \rangle + \langle U \rangle$. Thus

$$\boxed{\langle E \rangle = \frac{1}{2} \langle U \rangle,} \quad (2.47)$$

just what we found for the binary orbit problem.

The virial theorem applies to a wide variety of systems, from an ideal gas to a cluster of galaxies. For instance, consider the case of a static star. In equilibrium a star must obey the virial theorem, implying that its total energy is negative, one-half of the total potential energy. Assuming that the star formed as a result of the gravitational collapse of a large

cloud (a nebula), the potential energy of the system must have changed from an initial value of nearly zero to its negative static value. This implies that the star must have lost energy in the process, meaning that gravitational energy must have been radiated into space during the collapse. Applications of the virial theorem will be described in more detail in later chapters.

SUGGESTED READING

General

Kuhn, Thomas S., *The Structure of Scientific Revolutions*, Third Edition, University of Chicago Press, Chicago, 1996.

Westfall, Richard S., *Never at Rest: A Biography of Isaac Newton*, Cambridge University Press, Cambridge, 1980.

Technical

Arya, Atam P., *Introduction to Classical Mechanics*, Second Edition, Prentice Hall, Upper Saddle River, NJ, 1998.

Clayton, Donald D., *Principles of Stellar Evolution and Nucleosynthesis*, University of Chicago Press, New York, 1983.

Fowles, Grant R., and Cassiday, George L., *Analytical Mechanics*, Seventh Edition, Thomson Brooks/Cole, Belmont, CA, 2005.

Marion, Jerry B., and Thornton, Stephen T., *Classical Dynamics of Particles and Systems*, Fourth Edition, Saunders College Publishing, Fort Worth, 1995.

PROBLEMS

- 2.1** Assume that a rectangular coordinate system has its origin at the center of an elliptical planetary orbit and that the coordinate system's x axis lies along the major axis of the ellipse. Show that the equation for the ellipse is given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

where a and b are the lengths of the semimajor axis and the semiminor axis, respectively.

- 2.2** Using the result of Problem 2.1, prove that the area of an ellipse is given by $A = \pi ab$.
- 2.3** (a) Beginning with Eq. (2.3) and Kepler's second law, derive general expressions for $\mathbf{v}_r$ and $\mathbf{v}_\theta$ for a mass m_1 in an elliptical orbit about a second mass m_2 . Your final answers should be functions of P , e , a , and θ only.
- (b) Using the expressions for $\mathbf{v}_r$ and $\mathbf{v}_\theta$ that you derived in part (a), verify Eq. (2.36) directly from $v^2 = v_r^2 + v_\theta^2$.
- 2.4** Derive Eq. (2.25) from the sum of the kinetic and potential energy terms for the masses m_1 and m_2 .

- 2.5** Derive Eq. (2.26) from the total angular momentum of the masses m_1 and m_2 .
- 2.6** (a) Assuming that the Sun interacts only with Jupiter, calculate the total orbital angular momentum of the Sun–Jupiter system. The semimajor axis of Jupiter’s orbit is $a = 5.2$ AU, its orbital eccentricity is $e = 0.048$, and its orbital period is $P = 11.86$ yr.
- (b) Estimate the contribution the Sun makes to the total orbital angular momentum of the Sun–Jupiter system. For simplicity, assume that the Sun’s orbital eccentricity is $e = 0$, rather than $e = 0.048$. *Hint:* First find the distance of the center of the Sun from the center of mass.
- (c) Making the approximation that the orbit of Jupiter is a perfect circle, estimate the contribution it makes to the total orbital angular momentum of the Sun–Jupiter system. Compare your answer with the difference between the two values found in parts (a) and (b).
- (d) Recall that the moment of inertia of a solid sphere of mass m and radius r is given by $I = \frac{2}{5}mr^2$ when the sphere spins on an axis passing through its center. Furthermore, its rotational angular momentum may be written as

$$L = I\omega,$$

where ω is the angular frequency measured in rad s^{-1} . Assuming (incorrectly) that both the Sun and Jupiter rotate as solid spheres, calculate approximate values for the rotational angular momenta of the Sun and Jupiter. Take the rotation periods of the Sun and Jupiter to be 26 days and 10 hours, respectively. The radius of the Sun is 6.96×10^8 m, and the radius of Jupiter is 6.9×10^7 m.

- (e) What part of the Sun–Jupiter system makes the largest contribution to the total angular momentum?
- 2.7** (a) Using data contained in Problem 2.6 and in the chapter, calculate the escape speed at the surface of Jupiter.
- (b) Calculate the escape speed from the Solar System, starting from Earth’s orbit. Assume that the Sun constitutes all of the mass of the Solar System.
- 2.8** (a) The Hubble Space Telescope is in a nearly circular orbit, approximately 610 km (380 miles) above the surface of Earth. Estimate its orbital period.
- (b) Communications and weather satellites are often placed in *geosynchronous* “parking” orbits above Earth. These are orbits where satellites can remain fixed above a specific point on the surface of Earth. At what altitude must these satellites be located?
- (c) Is it possible for a satellite in a geosynchronous orbit to remain “parked” over any location on the surface of Earth? Why or why not?
- 2.9** In general, an *integral average* of some continuous function $f(t)$ over an interval τ is given by

$$\langle f(t) \rangle = \frac{1}{\tau} \int_0^\tau f(t) dt.$$

Beginning with an expression for the integral average, prove that

$$\langle U \rangle = -G \frac{M\mu}{a},$$

a binary system’s gravitational potential energy, averaged over one period, equals the value of the instantaneous potential energy of the system when the two masses are separated by the

distance a , the semimajor axis of the orbit of the reduced mass about the center of mass. *Hint:* You may find the following definite integral useful:

$$\int_0^{2\pi} \frac{d\theta}{1 + e \cos \theta} = \frac{2\pi}{\sqrt{1 - e^2}}.$$

- 2.10** Using the definition of the integral average given in Problem 2.9, prove that

$$\langle r \rangle \neq a$$

for the orbit of the reduced mass about the center of mass.

- 2.11** Given that a geocentric universe is (mathematically) only a matter of the choice of a reference frame, explain why the Ptolemaic model of the universe was able to survive scrutiny for such a long period of time.
- 2.12** Verify that Kepler's third law in the form of Eq. (2.37) applies to the four moons that Galileo discovered orbiting Jupiter (the Galilean moons: Io, Europa, Ganymede, and Callisto).
- (a) Using the data available in Appendix C, create a graph of $\log_{10} P$ vs. $\log_{10} a$.
 - (b) From the graph, show that the slope of the best-fit straight line through the data is $3/2$.
 - (c) Calculate the mass of Jupiter from the value of the y -intercept.
- 2.13** An alternative derivation of the total orbital angular momentum can be obtained by applying the conservation laws of angular momentum and energy.
- (a) From conservation of angular momentum, show that the ratio of orbital speeds at perihelion and aphelion is given by

$$\frac{v_p}{v_a} = \frac{1 + e}{1 - e}.$$

- (b) By equating the orbital mechanical energies at perihelion and aphelion, derive Eqs. (2.33) and (2.34) for the perihelion and aphelion speeds, respectively.
 - (c) Obtain Eq. (2.30) directly from the expression for v_p (or v_a).
- 2.14** Cometary orbits usually have very large eccentricities, often approaching (or even exceeding) unity. Halley's comet has an orbital period of 76 yr and an orbital eccentricity of $e = 0.9673$.
- (a) Using Kepler's original formulation of his third law as given on p. 25, determine the semimajor axis of Comet Halley's orbit.
 - (b) Using Newton's formulation of Kepler's third law (Eq. 2.37), use the orbital data of Comet Halley to estimate the mass of the Sun.
 - (c) Calculate the distance of Comet Halley from the Sun at perihelion and aphelion.
 - (d) Determine the orbital speed of the comet when at perihelion, at aphelion, and on the semiminor axis of its orbit.
 - (e) How many times larger is the kinetic energy of Halley's comet at perihelion than at aphelion?

COMPUTER PROBLEMS

- 2.15** Using *Orbit* (the program is described in Appendix J and available on the companion website) together with the data given in Problem 2.14, estimate the amount of time required for Halley's comet to move from perihelion to a distance of 1 AU away from the principal focus.

2.16 The code `Orbit` (Appendix J) can be used to generate orbital positions, given the mass of the central star, the semimajor axis of the orbit, and the orbital eccentricity. Using `Orbit` to generate the data, plot the orbits for three hypothetical objects orbiting our Sun. Assume that the semimajor axis of each orbit is 1 AU and that the orbital eccentricities are:

- (a) 0.0.
- (b) 0.4.
- (c) 0.9.

Note: Plot all three orbits on a common coordinate system and indicate the principal focus, located at $x = 0.0$, $y = 0.0$.

2.17 (a) From the data given in Example 2.1.1, use `Orbit` (Appendix J) to generate an orbit for Mars. Plot at least 25 points, evenly spaced in time, on a sheet of graph paper and clearly indicate the principal focus.

(b) Using a compass, draw a perfect circle on top of the elliptical orbit for Mars, choosing the radius of the circle and its center carefully in order to make the best possible approximation of the orbit. Be sure to mark the center of the circle you chose (note that it will not correspond to the principal focus of the elliptical orbit).

(c) What can you conclude about the merit of Kepler's first attempts to use offset circles and equants to model the orbit of Mars?

2.18 Figure 1.7 was drawn assuming perfectly circular motion and constant orbital speeds for Earth and Mars. By making very slight modifications to `Orbit` (Appendix J), a more realistic diagram can be created.

(a) Begin by assuming that Mars is initially at opposition and that Earth and Mars happen to be at their closest possible approach (aphelion and perihelion, respectively). Use your modified version of `Orbit` to calculate the positions of Earth and Mars between two successive oppositions of Mars. Graph the results.

(b) How much time (in years) elapsed between the two oppositions?

(c) Does your answer in part (b) agree precisely with the results obtained from Eq. (1.1)? Why or why not?

(d) Would you have obtained the same answer to part (b) if you had started the calculation with Earth at perihelion and Mars at aphelion? Explain your answer.

(e) From the results of your numerical experiment, explain why Mars appears brighter in the night sky during certain oppositions than during others.

CHAPTER

3

The Continuous Spectrum of Light

- 3.1 *Stellar Parallax*
- 3.2 *The Magnitude Scale*
- 3.3 *The Wave Nature of Light*
- 3.4 *Blackbody Radiation*
- 3.5 *The Quantization of Energy*
- 3.6 *The Color Index*

3.1 ■ STELLAR PARALLAX

Measuring the intrinsic brightness of stars is inextricably linked with determining their distances. This chapter on the light emitted by stars therefore begins with the problem of finding the distance to astronomical objects, one of the most important and most difficult tasks faced by astronomers. Kepler's laws in their original form describe the *relative* sizes of the planets' orbits in terms of astronomical units; their actual dimensions were unknown to Kepler and his contemporaries. The true scale of the Solar System was first revealed in 1761 when the distance to Venus was measured as it crossed the disk of the Sun in a rare transit during inferior conjunction. The method used was **trigonometric parallax**, the familiar surveyor's technique of triangulation. On Earth, the distance to the peak of a remote mountain can be determined by measuring that peak's angular position from two observation points separated by a known baseline distance. Simple trigonometry then supplies the distance to the peak; see Fig. 3.1. Similarly, the distances to the planets can be measured from two widely separated observation sites on Earth.

Finding the distance even to the nearest stars requires a longer baseline than Earth's diameter. As Earth orbits the Sun, two observations of the same star made 6 months apart employ a baseline equal to the diameter of Earth's orbit. These measurements reveal that a nearby star exhibits an annual back-and-forth change in its position against the stationary background of much more distant stars. (As mentioned in Section 1.3, a star may also change its position as a consequence of its own motion through space. However, this *proper motion*, seen from Earth, is not periodic and so can be distinguished from the star's periodic displacement caused by Earth's orbital motion.) As shown in Fig. 3.2, a measurement of the **parallax angle** p (one-half of the maximum change in angular position) allows the calculation of the distance d to the star.

$$d = \frac{1 \text{ AU}}{\tan p} \simeq \frac{1}{p} \text{ AU},$$

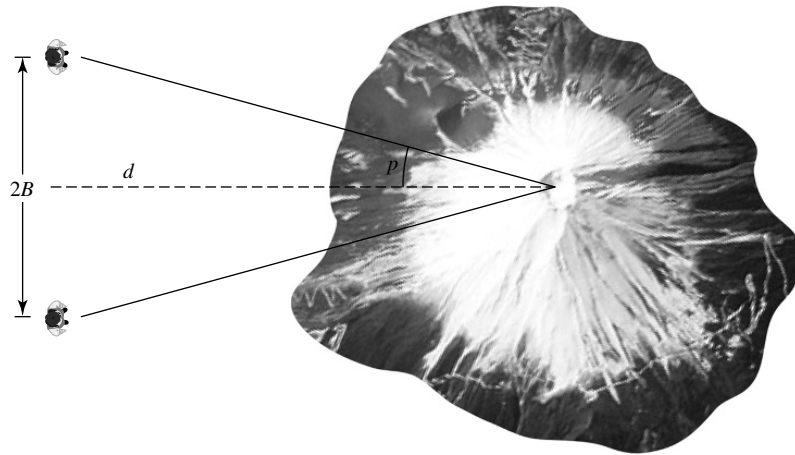


FIGURE 3.1 Trigonometric parallax: $d = B / \tan p$.

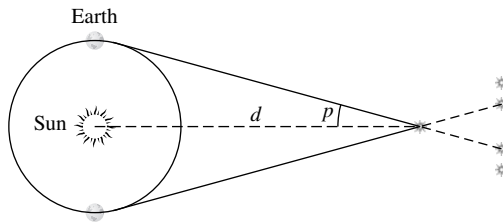


FIGURE 3.2 Stellar parallax: $d = 1/p''$ pc.

where the small-angle approximation $\tan p \simeq p$ has been employed for the parallax angle p measured in *radians*. Using $1 \text{ radian} = 57.2957795^\circ = 206264.806''$ to convert p to p'' in units of *arcseconds* produces

$$d \simeq \frac{206,265}{p''} \text{ AU}.$$

Defining a new unit of distance, the **parsec** (**parallax-second**, abbreviated pc), as $1 \text{ pc} = 2.06264806 \times 10^5 \text{ AU} = 3.0856776 \times 10^{16} \text{ m}$ leads to

$$d = \frac{1}{p''} \text{ pc.} \quad (3.1)$$

By definition, when the parallax angle $p = 1''$, the distance to the star is 1 pc. Thus 1 parsec is the distance from which the radius of Earth's orbit, 1 AU, subtends an angle of $1''$. Another unit of distance often encountered is the **light-year** (abbreviated ly), the distance traveled by light through a vacuum in one Julian year: $1 \text{ ly} = 9.460730472 \times 10^{15} \text{ m}$. One parsec is equivalent to 3.2615638 ly.

Even Proxima Centauri, the nearest star other than the Sun, has a parallax angle of less than $1''$. (Proxima Centauri is a member of the triple star system α Centauri, and has a parallax angle of $0.77''$. If Earth's orbit around the Sun were represented by a dime, then Proxima Centauri would be located 2.4 km away!) In fact, this cyclic change in a star's position is so difficult to detect that it was not until 1838 that it was first measured, by Friedrich Wilhelm Bessel (1784–1846), a German mathematician and astronomer.¹

Example 3.1.1. In 1838, after 4 years of observing 61 Cygni, Bessel announced his measurement of a parallax angle of $0.316''$ for that star. This corresponds to a distance of

$$d = \frac{1}{p''} \text{ pc} = \frac{1}{0.316} \text{ pc} = 3.16 \text{ pc} = 10.3 \text{ ly},$$

within 10% of the modern value 3.48 pc. 61 Cygni is one of the Sun's nearest neighbors.

From 1989 to 1993, the European Space Agency's (ESA's) Hipparcos Space Astrometry Mission operated high above Earth's distorting atmosphere.² The spacecraft was able to measure parallax angles with accuracies approaching $0.001''$ for over 118,000 stars, corresponding to a distance of $1000 \text{ pc} \equiv 1 \text{ kpc}$ (kiloparsec). Along with the high-precision Hipparcos experiment aboard the spacecraft, the lower-precision Tycho experiment produced a catalog of more than 1 million stars with parallaxes down to $0.02'' - 0.03''$. The two catalogs were published in 1997 and are available on CD-ROMs and the World Wide Web. Despite the impressive accuracy of the Hipparcos mission, the distances that were obtained are still quite small compared to the 8-kpc distance to the center of our Milky Way Galaxy, so stellar trigonometric parallax is currently useful only for surveying the local neighborhood of the Sun.

However, within the next decade, NASA plans to launch the Space Interferometry Mission (SIM PlanetQuest). This observatory will be able to determine positions, distances, and proper motions of stars with parallax angles as small as 4 *microarcseconds* ($0.000004''$), leading to the direct determination of distances of objects up to 250 kpc away, assuming that the objects are bright enough. In addition, ESA will launch the *Gaia* mission within the next decade as well, which will catalog the brightest 1 billion stars with parallax angles as small as 10 microarcseconds. With the anticipated levels of accuracy, these missions will be able to catalog stars and other objects across the Milky Way Galaxy and even in nearby galaxies. Clearly these ambitious projects will provide an amazing wealth of new information about the three-dimensional structure of our Galaxy and the nature of its constituents.

¹Tycho Brahe had searched for stellar parallax 250 years earlier, but his instruments were too imprecise to find it. Tycho concluded that Earth does not move through space, and he was thus unable to accept Copernicus's model of a heliocentric Solar System.

²**Astrometry** is the subdiscipline of astronomy that is concerned with the three-dimensional positions of celestial objects.

3.2 ■ THE MAGNITUDE SCALE

Nearly all of the information astronomers have received about the universe beyond our Solar System has come from the careful study of the light emitted by stars, galaxies, and interstellar clouds of gas and dust. Our modern understanding of the universe has been made possible by the quantitative measurement of the intensity and polarization of light in every part of the electromagnetic spectrum.

Apparent Magnitude

The Greek astronomer Hipparchus was one of the first sky watchers to catalog the stars that he saw. In addition to compiling a list of the positions of some 850 stars, Hipparchus invented a numerical scale to describe how bright each star appeared in the sky. He assigned an **apparent magnitude** $m = 1$ to the brightest stars in the sky, and he gave the dimmest stars visible to the naked eye an apparent magnitude of $m = 6$. Note that a smaller apparent magnitude means a brighter-appearing star.

Since Hipparchus's time, astronomers have extended and refined his apparent magnitude scale. In the nineteenth century, it was thought that the human eye responded to the difference in the *logarithms* of the brightness of two luminous objects. This theory led to a scale in which a difference of 1 magnitude between two stars implies a constant *ratio* between their brightnesses. By the modern definition, a difference of 5 magnitudes corresponds exactly to a factor of 100 in brightness, so a difference of 1 magnitude corresponds exactly to a brightness ratio of $100^{1/5} \simeq 2.512$. Thus a first-magnitude star appears 2.512 times brighter than a second-magnitude star, $2.512^2 = 6.310$ times brighter than a third-magnitude star, and 100 times brighter than a sixth-magnitude star.

Using sensitive detectors, astronomers can measure the apparent magnitude of an object with an accuracy of ± 0.01 magnitude, and *differences* in magnitudes with an accuracy of ± 0.002 magnitude. Hipparchus's scale has been extended in both directions, from $m = -26.83$ for the Sun to approximately $m = 30$ for the faintest object detectable.³ The total range of nearly 57 magnitudes corresponds to over $100^{57/5} = (10^2)^{11.4} \simeq 10^{23}$ for the ratio of the apparent brightness of the Sun to that of the faintest star or galaxy yet observed.

Flux, Luminosity, and the Inverse Square Law

The “brightness” of a star is actually measured in terms of the **radiant flux** F received from the star. The radiant flux is the total amount of light energy of all wavelengths that crosses a unit area oriented perpendicular to the direction of the light's travel per unit time; that is, it is the number of joules of starlight energy per second (i.e., the number of watts) received by one square meter of a detector aimed at the star. Of course, the radiant flux received from an object depends on both its intrinsic **luminosity** (energy emitted per second) and its distance from the observer. The same star, if located farther from Earth, would appear less bright in the sky.

³The magnitudes discussed in this section are actually *bolometric* magnitudes, measured over all wavelengths of light; see Section 3.6 for a discussion of magnitudes measured by detectors over a finite wavelength region.

Imagine a star of luminosity L surrounded by a huge spherical shell of radius r . Then, assuming that no light is absorbed during its journey out to the shell, the radiant flux, F , measured at distance r is related to the star's luminosity by

$$F = \frac{L}{4\pi r^2}, \quad (3.2)$$

the denominator being simply the area of the sphere. Since L does not depend on r , the radiant flux is inversely proportional to the square of the distance from the star. This is the well-known **inverse square law** for light.⁴

Example 3.2.1. The luminosity of the Sun is $L_{\odot} = 3.839 \times 10^{26}$ W. At a distance of 1 AU = 1.496×10^{11} m, Earth receives a radiant flux above its absorbing atmosphere of

$$F = \frac{L}{4\pi r^2} = 1365 \text{ W m}^{-2}.$$

This value of the solar flux is known as the **solar irradiance**, sometimes also called the **solar constant**. At a distance of 10 pc = 2.063×10^6 AU, an observer would measure the radiant flux to be only $1/(2.063 \times 10^6)^2$ as large. That is, the radiant flux from the Sun would be 3.208×10^{-10} W m⁻² at a distance of 10 pc.

Absolute Magnitude

Using the inverse square law, astronomers can assign an **absolute magnitude**, M , to each star. This is defined to be the apparent magnitude a star would have *if* it were located at a distance of 10 pc. Recall that a difference of 5 magnitudes between the apparent magnitudes of two stars corresponds to the smaller-magnitude star being 100 times brighter than the larger-magnitude star. This allows us to specify their flux ratio as

$$\frac{F_2}{F_1} = 100^{(m_1 - m_2)/5}. \quad (3.3)$$

Taking the logarithm of both sides leads to the alternative form:

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right). \quad (3.4)$$

The Distance Modulus

The connection between a star's apparent and absolute magnitudes and its distance may be found by combining Eqs. (3.2) and (3.3):

$$100^{(m-M)/5} = \frac{F_{10}}{F} = \left(\frac{d}{10 \text{ pc}} \right)^2,$$

⁴If the star is moving with a speed near that of light, the inverse square law must be modified slightly.

where F_{10} is the flux that would be received if the star were at a distance of 10 pc, and d is the star's distance, measured in *parsecs*. Solving for d gives

$$d = 10^{(m-M+5)/5} \text{ pc.} \quad (3.5)$$

The quantity $m - M$ is therefore a measure of the distance to a star and is called the star's **distance modulus**:

$$m - M = 5 \log_{10}(d) - 5 = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right). \quad (3.6)$$

Example 3.2.2. The apparent magnitude of the Sun is $m_{\text{Sun}} = -26.83$, and its distance is 1 AU = 4.848×10^{-6} pc. Equation (3.6) shows that the absolute magnitude of the Sun is

$$M_{\text{Sun}} = m_{\text{Sun}} - 5 \log_{10}(d) + 5 = +4.74,$$

as already given. The Sun's distance modulus is thus $m_{\text{Sun}} - M_{\text{Sun}} = -31.57$.⁵

For two stars at the same distance, Eq. (3.2) shows that the ratio of their radiant fluxes is equal to the ratio of their luminosities. Thus Eq. (3.3) for absolute magnitudes becomes

$$100^{(M_1 - M_2)/5} = \frac{L_2}{L_1}. \quad (3.7)$$

Letting one of these stars be the Sun reveals the direct relation between a star's absolute magnitude and its luminosity:

$$M = M_{\text{Sun}} - 2.5 \log_{10} \left(\frac{L}{L_{\odot}} \right), \quad (3.8)$$

where the absolute magnitude and luminosity of the Sun are $M_{\text{Sun}} = +4.74$ and $L_{\odot} = 3.839 \times 10^{26}$ W, respectively. It is left as an exercise for you to show that a star's apparent magnitude m is related to the radiant flux F received from the star by

$$m = M_{\text{Sun}} - 2.5 \log_{10} \left(\frac{F}{F_{10, \odot}} \right), \quad (3.9)$$

where $F_{10, \odot}$ is the radiant flux received from the Sun at a distance of 10 pc (see Example 3.2.1).

The inverse square law for light, Eq. (3.2), relates the intrinsic properties of a star (luminosity L and absolute magnitude M) to the quantities measured at a distance from

⁵The magnitudes m and M for the Sun have a “Sun” subscript (instead of “ $\odot$ ”) to avoid confusion with $M_{\odot}$, the standard symbol for the Sun's mass.

that star (radiant flux F and apparent magnitude m). At first glance, it may seem that astronomers must start with the measurable quantities F and m and then use the distance to the star (if known) to determine the star's intrinsic properties. However, if the star belongs to an important class of objects known as *pulsating variable stars*, its intrinsic luminosity L and absolute magnitude M can be determined *without* any knowledge of its distance. Equation (3.5) then gives the distance to the variable star. As will be discussed in Section 14.1, these stars act as beacons that illuminate the fundamental distance scale of the universe.

3.3 ■ THE WAVE NATURE OF LIGHT

Much of the history of physics is concerned with the evolution of our ideas about the nature of light.

The Speed of Light

The speed of light was first measured with some accuracy in 1675, by the Danish astronomer Ole Roemer (1644–1710). Roemer observed the moons of Jupiter as they passed into the giant planet's shadow, and he was able to calculate when future eclipses of the moons should occur by using Kepler's laws. However, Roemer discovered that when Earth was moving closer to Jupiter, the eclipses occurred earlier than expected. Similarly, when Earth was moving away from Jupiter, the eclipses occurred behind schedule. Roemer realized that the discrepancy was caused by the differing amounts of time it took for light to travel the changing distance between the two planets, and he concluded that 22 minutes was required for light to cross the diameter of Earth's orbit.⁶ The resulting value of $2.2 \times 10^8 \text{ m s}^{-1}$ was close to the modern value of the speed of light. In 1983 the speed of light *in vacuo* was formally *defined* to be $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$, and the unit of length (the meter) is now derived from this value.⁷

Young's Double-Slit Experiment

Even the fundamental nature of light has long been debated. Isaac Newton, for example, believed that light must consist of a rectilinear stream of particles, because only such a stream could account for the sharpness of shadows. Christian Huygens (1629–1695), a contemporary of Newton, advanced the idea that light must consist of waves. According to Huygens, light is described by the usual quantities appropriate for a wave. The distance between two successive wave crests is the **wavelength** λ , and the number of waves per second that pass a point in space is the **frequency** ν of the wave. Then the speed of the light wave is given by

$$c = \lambda \nu. \quad (3.10)$$

⁶We now know that it takes light about 16.5 minutes to travel 2 AU.

⁷In 1905 Albert Einstein realized that the speed of light is a universal constant of nature whose value is independent of the observer (see page 88). This realization plays a central role in his Special Theory of Relativity (Chapter 4).

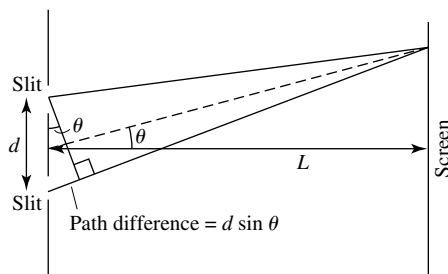


FIGURE 3.3 Double-slit experiment.

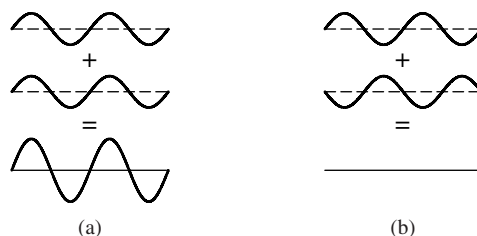


FIGURE 3.4 Superposition principle for light waves. (a) Constructive interference. (b) Destructive interference.

Both the particle and wave models could explain the familiar phenomena of the reflection and refraction of light. However, the particle model prevailed, primarily on the strength of Newton's reputation, until the wave nature of light was conclusively demonstrated by Thomas Young's (1773–1829) famous double-slit experiment.

In a double-slit experiment, monochromatic light of wavelength λ from a single source passes through two narrow, parallel slits that are separated by a distance d . The light then falls upon a screen a distance L beyond the two slits (see Fig. 3.3). The series of light and dark *interference fringes* that Young observed on the screen could be explained only by a wave model of light. As the light waves pass through the narrow slits,⁸ they spread out (diffract) radially in a succession of crests and troughs. Light obeys a *superposition principle*, so when two waves meet, they add algebraically; see Fig. 3.4. At the screen, if a wave crest from one slit meets a wave crest from the other slit, a bright fringe or maximum is produced by the resulting **constructive interference**. But if a wave crest from one slit meets a wave trough from the other slit, they cancel each other, and a dark fringe or minimum results from this **destructive interference**.

The interference pattern observed thus depends on the difference in the lengths of the paths traveled by the light waves from the two slits to the screen. As shown in Fig. 3.3, if $L \gg d$, then to a good approximation this path difference is $d \sin \theta$. The light waves will arrive at the screen *in phase* if the path difference is equal to an integral number of wavelengths. On the other hand, the light waves will arrive 180° *out of phase* if the path difference is equal to an odd integral number of half-wavelengths. So for $L \gg d$, the angular

⁸Actually, Young used pinholes in his original experiment.

positions of the bright and dark fringes for **double-slit interference** are given by

$$d \sin \theta = \begin{cases} n\lambda & (n = 0, 1, 2, \dots \text{ for bright fringes}) \\ (n - \frac{1}{2})\lambda & (n = 1, 2, 3, \dots \text{ for dark fringes}). \end{cases} \quad (3.11)$$

In either case, n is called the **order** of the maximum or minimum. From the measured positions of the light and dark fringes on the screen, Young was able to determine the wavelength of the light. At the short-wavelength end, Young found that violet light has a wavelength of approximately 400 nm, while at the long-wavelength end, red light has a wavelength of only 700 nm.⁹ The diffraction of light goes unnoticed under everyday conditions for these short wavelengths, thus explaining Newton's sharp shadows.

Maxwell's Electromagnetic Wave Theory

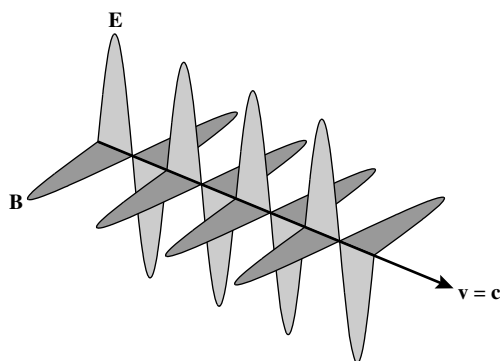
The nature of these waves of light remained elusive until the early 1860s, when the Scottish mathematical physicist James Clerk Maxwell (1831–1879) succeeded in condensing everything known about electric and magnetic fields into the four equations that today bear his name. Maxwell found that his equations could be manipulated to produce wave equations for the electric and magnetic field vectors **E** and **B**. These wave equations predicted the existence of *electromagnetic waves* that travel through a vacuum with speed $v = 1/\sqrt{\epsilon_0\mu_0}$, where ϵ_0 and μ_0 are fundamental constants associated with the electric and magnetic fields, respectively. Upon inserting the values of ϵ_0 and μ_0 , Maxwell was amazed to discover that electromagnetic waves travel at the speed of light. Furthermore, these equations implied that electromagnetic waves are *transverse* waves, with the oscillatory electric and magnetic fields perpendicular to each other *and* to the direction of the wave's propagation (see Fig. 3.5); such waves could exhibit the polarization¹⁰ known to occur for light. Maxwell wrote that “we can scarcely avoid the inference that light consists in the transverse modulations of the same medium which is the cause of electric and magnetic phenomena.”

Maxwell did not live to see the experimental verification of his prediction of electromagnetic waves. Ten years after Maxwell's death, the German physicist Heinrich Hertz (1857–1894) succeeded in producing radio waves in his laboratory. Hertz determined that these electromagnetic waves do indeed travel at the speed of light, and he confirmed their reflection, refraction, and polarization properties. In 1889, Hertz wrote:

What is light? Since the time of Young and Fresnel we know that it is wave motion. We know the velocity of the waves, we know their lengths, and we know that they are transverse; in short, our knowledge of the geometrical conditions of the motion is complete. A doubt about these things is no longer possible; a refutation of these views is inconceivable to the physicist. The wave theory of light is, from the point of view of human beings, certainty.

⁹Another commonly used measure of the wavelength of light is the **angstrom**; $1 \text{ \AA} = 0.1 \text{ nm}$. In these units, violet light has a wavelength of 4000 Å and red light has a wavelength of 7000 Å.

¹⁰The electromagnetic wave shown in Fig. 3.5 is *plane-polarized*, with its electric and magnetic fields oscillating in planes. Because **E** and **B** are always perpendicular, their respective planes of polarization are perpendicular as well.

**FIGURE 3.5** Electromagnetic wave.**TABLE 3.1** The Electromagnetic Spectrum.

Region	Wavelength
Gamma ray	$\lambda < 0.01 \text{ nm}$
X-ray	$0.01 \text{ nm} < \lambda < 10 \text{ nm}$
Ultraviolet	$10 \text{ nm} < \lambda < 400 \text{ nm}$
Visible	$400 \text{ nm} < \lambda < 700 \text{ nm}$
Infrared	$700 \text{ nm} < \lambda < 1 \text{ mm}$
Microwave	$1 \text{ mm} < \lambda < 10 \text{ cm}$
Radio	$10 \text{ cm} < \lambda$

The Electromagnetic Spectrum

Today, astronomers utilize light from every part of the **electromagnetic spectrum**. The total spectrum of light consists of electromagnetic waves of all wavelengths, ranging from very short-wavelength gamma rays to very long-wavelength radio waves. Table 3.1 shows how the electromagnetic spectrum has been arbitrarily divided into various wavelength regions.

The Poynting Vector and Radiation Pressure

Like all waves, electromagnetic waves carry both energy and momentum in the direction of propagation. The rate at which energy is carried by a light wave is described by the **Poynting vector**,¹¹

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B},$$

where $\mathbf{S}$ has units of W m^{-2} . The Poynting vector points in the direction of the electromagnetic wave's propagation and has a magnitude equal to the amount of energy per unit time that crosses a unit area oriented perpendicular to the direction of the propagation of

¹¹The Poynting vector is named after John Henry Poynting (1852–1914), the physicist who first described it.

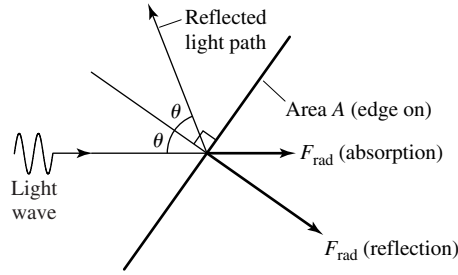


FIGURE 3.6 Radiation pressure force. The surface area A is seen edge on.

the wave. Because the magnitudes of the fields $\mathbf{E}$ and $\mathbf{B}$ vary harmonically with time, the quantity of practical interest is the *time-averaged* value of the Poynting vector over one cycle of the electromagnetic wave. In a vacuum the magnitude of the time-averaged Poynting vector, $\langle S \rangle$, is

$$\langle S \rangle = \frac{1}{2\mu_0} E_0 B_0, \quad (3.12)$$

where E_0 and B_0 are the maximum magnitudes (amplitudes) of the electric and magnetic fields, respectively. (For an electromagnetic wave in a vacuum, E_0 and B_0 are related by $E_0 = cB_0$.) The time-averaged Poynting vector thus provides a description of the radiant flux in terms of the electric and magnetic fields of the light waves. However, it should be remembered that the radiant flux discussed in Section 3.2 involves the amount of energy received *at all wavelengths* from a star, whereas E_0 and B_0 describe an electromagnetic wave of a specified wavelength.

Because an electromagnetic wave carries momentum, it can exert a force on a surface hit by the light. The resulting **radiation pressure** depends on whether the light is reflected from or absorbed by the surface. Referring to Fig. 3.6, if the light is completely absorbed, then the force due to radiation pressure is in the direction of the light's propagation and has magnitude

$$F_{\text{rad}} = \frac{\langle S \rangle A}{c} \cos \theta \quad (\text{absorption}), \quad (3.13)$$

where θ is the angle of incidence of the light as measured from the direction perpendicular to the surface of area A . Alternatively, if the light is completely reflected, then the radiation pressure force must act in a direction perpendicular to the surface; the reflected light cannot exert a force parallel to the surface. Then the magnitude of the force is

$$F_{\text{rad}} = \frac{2\langle S \rangle A}{c} \cos^2 \theta \quad (\text{reflection}). \quad (3.14)$$

Radiation pressure has a negligible effect on physical systems under everyday conditions. However, radiation pressure may play a dominant role in determining some aspects of the behavior of extremely luminous objects such as early main-sequence stars, red supergiants, and accreting compact stars. It may also have a significant effect on the small particles of dust found throughout the interstellar medium.

3.4 ■ BLACKBODY RADIATION

Anyone who has looked at the constellation of Orion on a clear winter night has noticed the strikingly different colors of red Betelgeuse (in Orion's northeast shoulder) and blue-white Rigel (in the southwest leg); see Fig. 3.7. These colors betray the difference in the surface temperatures of the two stars. Betelgeuse has a surface temperature of roughly 3600 K, significantly cooler than the 13,000-K surface of Rigel.¹²

The Connection between Color and Temperature

The connection between the color of light emitted by a hot object and its temperature was first noticed in 1792 by the English maker of fine porcelain Thomas Wedgewood. All of his ovens became red-hot at the same temperature, independent of their size, shape, and construction. Subsequent investigations by many physicists revealed that any object with a temperature above absolute zero emits light of all wavelengths with varying degrees of efficiency; an *ideal emitter* is an object that absorbs *all* of the light energy incident upon it and reradiates this energy with the characteristic spectrum shown in Fig. 3.8. Because an ideal emitter reflects no light, it is known as a **blackbody**, and the radiation it emits is called **blackbody radiation**. Stars and planets are blackbodies, at least to a rough first approximation.

Figure 3.8 shows that a blackbody of temperature T emits a **continuous spectrum** with some energy at all wavelengths and that this blackbody spectrum peaks at a wavelength λ_{max} , which becomes shorter with increasing temperature. The relation between λ_{max} and

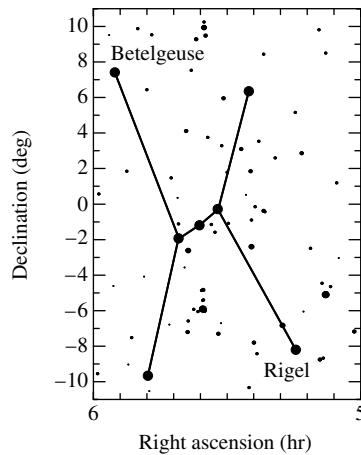


FIGURE 3.7 The constellation of Orion.

¹²Both of these stars are pulsating variables (Chapter 14), so the values quoted are *average* temperatures. Estimates of the surface temperature of Betelgeuse actually range quite widely, from about 3100 K to 3900 K. Similarly, estimates of the surface temperature of Rigel range from 8000 K to 13,000 K.

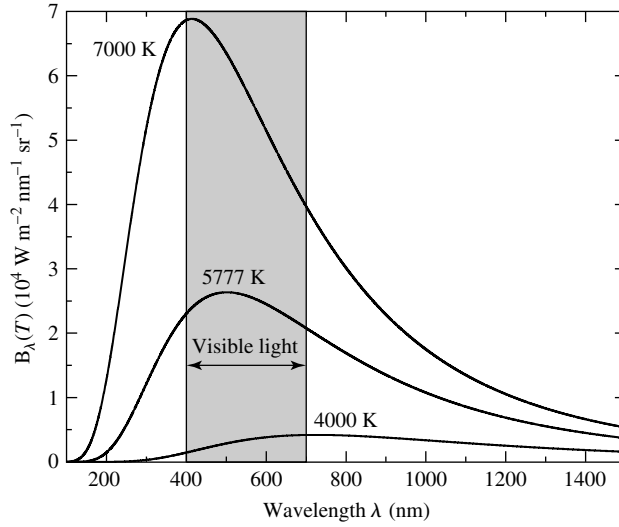


FIGURE 3.8 Blackbody spectrum [Planck function $B_\lambda(T)$].

T is known as **Wien's displacement law**.¹³

$$\lambda_{\max} T = 0.002897755 \text{ m K}. \quad (3.15)$$

Example 3.4.1. Betelgeuse has a surface temperature of 3600 K. If we treat Betelgeuse as a blackbody, Wien's displacement law shows that its continuous spectrum peaks at a wavelength of

$$\lambda_{\max} \simeq \frac{0.0029 \text{ m K}}{3600 \text{ K}} = 8.05 \times 10^{-7} \text{ m} = 805 \text{ nm},$$

which is in the infrared region of the electromagnetic spectrum. Rigel, with a surface temperature of 13,000 K, has a continuous spectrum that peaks at a wavelength of

$$\lambda_{\max} \simeq \frac{0.0029 \text{ m K}}{13,000 \text{ K}} = 2.23 \times 10^{-7} \text{ m} = 223 \text{ nm},$$

in the ultraviolet region.

The Stefan–Boltzmann Equation

Figure 3.8 also shows that as the temperature of a blackbody increases, it emits more energy per second at *all* wavelengths. Experiments performed by the Austrian physicist

¹³In 1911, the German physicist Wilhelm Wien (1864–1928) received the Nobel Prize in 1911 for his theoretical contributions to understanding the blackbody spectrum.

Josef Stefan (1835–1893) in 1879 showed that the luminosity, L , of a blackbody of area A and temperature T (in kelvins) is given by

$$L = A\sigma T^4. \quad (3.16)$$

Five years later another Austrian physicist, Ludwig Boltzmann (1844–1906), derived this equation, now called the **Stefan–Boltzmann equation**, using the laws of thermodynamics and Maxwell’s formula for radiation pressure. The Stefan–Boltzmann constant, σ , has the value

$$\sigma = 5.670400 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

For a spherical star of radius R and surface area $A = 4\pi R^2$, the Stefan–Boltzmann equation takes the form

$$L = 4\pi R^2 \sigma T_e^4. \quad (3.17)$$

Since stars are not perfect blackbodies, we use this equation to *define* the **effective temperature** T_e of a star’s surface. Combining this with the inverse square law, Eq. (3.2), shows that at the surface of the star ($r = R$), the *surface flux* is

$$F_{\text{surf}} = \sigma T_e^4. \quad (3.18)$$

Example 3.4.2. The luminosity of the Sun is $L_{\odot} = 3.839 \times 10^{26} \text{ W}$ and its radius is $R_{\odot} = 6.95508 \times 10^8 \text{ m}$. The effective temperature of the Sun’s surface is then

$$T_{\odot} = \left(\frac{L_{\odot}}{4\pi R_{\odot}^2 \sigma} \right)^{\frac{1}{4}} = 5777 \text{ K}.$$

The radiant flux at the solar surface is

$$F_{\text{surf}} = \sigma T_{\odot}^4 = 6.316 \times 10^7 \text{ W m}^{-2}.$$

According to Wien’s displacement law, the Sun’s continuous spectrum peaks at a wavelength of

$$\lambda_{\text{max}} \simeq \frac{0.0029 \text{ m K}}{5777 \text{ K}} = 5.016 \times 10^{-7} \text{ m} = 501.6 \text{ nm}.$$

This wavelength falls in the *green* region ($491 \text{ nm} < \lambda < 575 \text{ nm}$) of the spectrum of visible light. However, the Sun emits a continuum of wavelengths both shorter and longer than λ_{max} , and the human eye perceives the Sun’s color as yellow. Because the Sun emits most of its energy at visible wavelengths (see Fig. 3.8), and because Earth’s atmosphere is transparent at these wavelengths, the evolutionary process of natural selection has produced a human eye sensitive to this wavelength region of the electromagnetic spectrum.

Rounding off λ_{max} and $T_{\odot}$ to the values of 500 nm and 5800 K, respectively, permits Wien’s displacement law to be written in the approximate form

$$\lambda_{\text{max}} T \approx (500 \text{ nm})(5800 \text{ K}). \quad (3.19)$$

The Eve of a New World View

This section draws to a close at the end of the nineteenth century. The physicists and astronomers of the time believed that all of the principles that govern the physical world had finally been discovered. Their scientific world view, the *Newtonian paradigm*, was the culmination of the heroic, golden age of classical physics that had flourished for over three hundred years. The construction of this paradigm began with the brilliant observations of Galileo and the subtle insights of Newton. Its architecture was framed by Newton's laws, supported by the twin pillars of the conservation of energy and momentum and illuminated by Maxwell's electromagnetic waves. Its legacy was a deterministic description of a universe that ran like clockwork, with wheels turning inside of wheels, all of its gears perfectly meshed. Physics was in danger of becoming a victim of its own success. There were no challenges remaining. All of the great discoveries apparently had been made, and the only task remaining for men and women of science at the end of the nineteenth century was filling in the details.

However, as the twentieth century opened, it became increasingly apparent that a crisis was brewing. Physicists were frustrated by their inability to answer some of the simplest questions concerning light. What is the medium through which light waves travel the vast distances between the stars, and what is Earth's speed through this medium? What determines the continuous spectrum of blackbody radiation and the characteristic, discrete colors of tubes filled with hot glowing gases? Astronomers were tantalized by hints of a treasure of knowledge just beyond their grasp.

It took a physicist of the stature of Albert Einstein to topple the Newtonian paradigm and bring about two revolutions in physics. One transformed our ideas about space and time, and the other changed our basic concepts of matter and energy. The rigid clockwork universe of the golden era was found to be an illusion and was replaced by a random universe governed by the laws of probability and statistics. The following four lines aptly summarize the situation. The first two lines were written by the English poet Alexander Pope (1688–1744), a contemporary of Newton; the last two, by Sir J. C. Squire (1884–1958), were penned in 1926.

Nature and Nature's laws lay hid in night:
God said, *Let Newton be!* and all was light.

It did not last: the Devil howling "Ho!"
Let Einstein be!" restored the status quo.

3.5 ■ THE QUANTIZATION OF ENERGY

One of the problems haunting physicists at the end of the nineteenth century was their inability to derive from fundamental physical principles the blackbody radiation curve depicted in Fig. 3.8. Lord Rayleigh¹⁴ (1842–1919) had attempted to arrive at the expression by applying Maxwell's equations of classical electromagnetic theory together with the results

¹⁴Lord Rayleigh, as he is commonly known, was born John William Strutt but succeeded to the title of third Baron Rayleigh of Terling Place, Witham, in the county of Essex, when he was thirty years old.

from thermal physics. His strategy was to consider a cavity of temperature T filled with blackbody radiation. This may be thought of as a hot oven filled with standing waves of electromagnetic radiation. If L is the distance between the oven's walls, then the permitted wavelengths of the radiation are $\lambda = 2L, L, 2L/3, 2L/4, 2L/5, \dots$, extending forever to increasingly shorter wavelengths.¹⁵ According to classical physics, each of these wavelengths should receive an amount of energy equal to kT , where $k = 1.3806503 \times 10^{-23} \text{ J K}^{-1}$ is Boltzmann's constant, familiar from the ideal gas law $PV = NkT$. The result of Rayleigh's derivation gave

$$B_\lambda(T) \simeq \frac{2ckT}{\lambda^4}, \quad (\text{valid only if } \lambda \text{ is long}) \quad (3.20)$$

which agrees well with the long-wavelength tail of the blackbody radiation curve. However, a severe problem with Rayleigh's result was recognized immediately; his solution for $B_\lambda(T)$ grows without limit as $\lambda \rightarrow 0$. The source of the problem is that according to classical physics, an infinite number of infinitesimally short wavelengths implied that an unlimited amount of blackbody radiation energy was contained in the oven, a theoretical result so absurd it was dubbed the "ultraviolet catastrophe." Equation (3.20) is known today as the **Rayleigh–Jeans law**.¹⁶

Wien was also working on developing the correct mathematical expression for the blackbody radiation curve. Guided by the Stefan–Boltzmann law (Eq. 3.16) and classical thermal physics, Wien was able to develop an empirical law that described the curve at short wavelengths but failed at longer wavelengths:

$$B_\lambda(T) \simeq a\lambda^{-5}e^{-b/\lambda T}, \quad (\text{valid only if } \lambda \text{ is short}) \quad (3.21)$$

where a and b were constants chosen to provide the best fit to the experimental data.

Planck's Function for the Blackbody Radiation Curve

By late 1900 the German physicist Max Planck (1858–1947) had discovered that a modification of Wien's expression could be made to fit the blackbody spectra shown in Fig. 3.8 while simultaneously replicating the long-wavelength success of the Rayleigh–Jeans law and avoiding the ultraviolet catastrophe:

$$B_\lambda(T) = \frac{a/\lambda^5}{e^{b/\lambda T} - 1},$$

In order to determine the constants a and b while circumventing the ultraviolet catastrophe, Planck employed a clever mathematical trick. He assumed that a standing electromagnetic wave of wavelength λ and frequency $\nu = c/\lambda$ could not acquire just any arbitrary amount of energy. Instead, the wave could have only specific allowed energy values that

¹⁵This is analogous to standing waves on a string of length L that is held fixed at both ends. The permitted wavelengths are the same as those of the standing electromagnetic waves.

¹⁶James Jeans (1877–1946), a British astronomer, found a numerical error in Rayleigh's original work; the corrected result now bears the names of both men.

were integral multiples of a minimum wave energy.¹⁷ This minimum energy, a **quantum** of energy, is given by $h\nu$ or hc/λ , where h is a constant. Thus the energy of an electromagnetic wave is $nh\nu$ or nhc/λ , where n (an integer) is the number of quanta in the wave. Given this assumption of quantized wave energy with a minimum energy proportional to the frequency of the wave, the entire oven could not contain enough energy to supply even one quantum of energy for the short-wavelength, high-frequency waves. Thus the ultraviolet catastrophe would be avoided. Planck hoped that at the end of his derivation, the constant h could be set to zero; certainly, an artificial constant should not remain in his final result for $B_\lambda(T)$.

Planck's stratagem worked! His formula, now known as the **Planck function**, agreed wonderfully with experiment, but *only* if the constant h remained in the equation:¹⁸

$$B_\lambda(T) = \frac{2hc^2/\lambda^5}{e^{hc/\lambda kT} - 1}. \quad (3.22)$$

Planck's constant has the value $h = 6.62606876 \times 10^{-34}$ J s.

The Planck Function and Astrophysics

Finally armed with the correct expression for the blackbody spectrum, we can apply Planck's function to astrophysical systems. In spherical coordinates, the amount of radiant energy per unit time having wavelengths between λ and $\lambda + d\lambda$ emitted by a blackbody of temperature T and surface area dA into a solid angle $d\Omega \equiv \sin \theta d\theta d\phi$ is given by

$$B_\lambda(T) d\lambda dA \cos \theta d\Omega = B_\lambda(T) d\lambda dA \cos \theta \sin \theta d\theta d\phi; \quad (3.23)$$

see Fig. 3.9.¹⁹ The units of B_λ are therefore $\text{W m}^{-3} \text{sr}^{-1}$. Unfortunately, these units can be misleading. You should note that " W m^{-3} " indicates power (energy per unit time) per unit area per unit wavelength interval, $\text{W m}^{-2} \text{m}^{-1}$, *not* energy per unit time per unit volume. To help avoid confusion, the units of the wavelength interval $d\lambda$ are sometimes expressed in nanometers rather than meters, so the units of the Planck function become $\text{W m}^{-2} \text{nm}^{-1} \text{sr}^{-1}$, as in Fig. 3.8.²⁰

At times it is more convenient to deal with frequency intervals $d\nu$ than with wavelength intervals $d\lambda$. In this case the Planck function has the form

$$B_\nu(T) = \frac{2h\nu^3/c^2}{e^{h\nu/kT} - 1}. \quad (3.24)$$

¹⁷Actually, Planck restricted the possible energies of hypothetical electromagnetic oscillators in the oven walls that emit the electromagnetic radiation.

¹⁸It is left for you to show that the Planck function reduces to the Rayleigh–Jeans law at long wavelengths (Problem 3.10) and to Wien's expression at short wavelengths (Problem 3.11).

¹⁹Note that $dA \cos \theta$ is the area dA projected onto a plane perpendicular to the direction in which the radiation is traveling. The concept of a solid angle will be fully described in Section 6.1.

²⁰The value of the Planck function thus depends on the units of the wavelength interval. The conversion of $d\lambda$ from meters to nanometers means that the values of B_λ obtained by evaluating Eq. (3.22) must be divided by 10^9 .

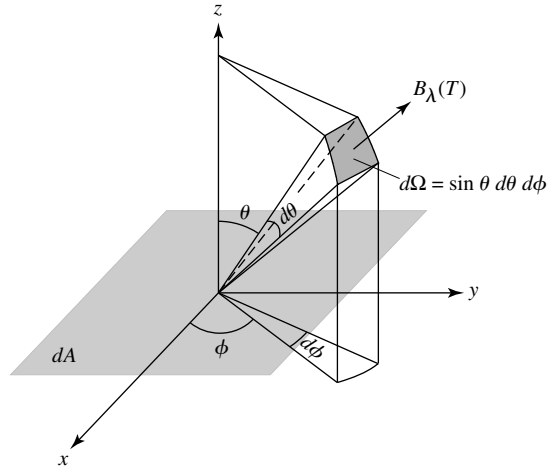


FIGURE 3.9 Blackbody radiation from an element of surface area dA .

Thus, in spherical coordinates,

$$B_\nu d\nu dA \cos \theta d\Omega = B_\nu d\nu dA \cos \theta \sin \theta d\theta d\phi$$

is the amount of energy per unit time of blackbody radiation having frequency between ν and $\nu + d\nu$ emitted by a blackbody of temperature T and surface area dA into a solid angle $d\Omega = \sin \theta d\theta d\phi$.

The Planck function can be used to make the connection between the observed properties of a star (radiant flux, apparent magnitude) and its intrinsic properties (radius, temperature). Consider a model star consisting of a spherical blackbody of radius R and temperature T . Assuming that each small patch of surface area dA emits blackbody radiation *isotropically* (equally in all directions) over the outward hemisphere, the energy per second having wavelengths between λ and $\lambda + d\lambda$ emitted by the star is

$$L_\lambda d\lambda = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \int_A B_\lambda d\lambda dA \cos \theta \sin \theta d\theta d\phi. \quad (3.25)$$

The angular integration yields a factor of π , and the integral over the area of the sphere produces a factor of $4\pi R^2$. The result is

$$L_\lambda d\lambda = 4\pi^2 R^2 B_\lambda d\lambda \quad (3.26)$$

$$= \frac{8\pi^2 R^2 hc^2 / \lambda^5}{e^{hc/\lambda kT} - 1} d\lambda. \quad (3.27)$$

$L_\lambda d\lambda$ is known as the **monochromatic luminosity**. Comparing the Stefan–Boltzmann equation (3.17) with the result of integrating Eq. (3.26) over all wavelengths shows that

$$\int_0^\infty B_\lambda(T) d\lambda = \frac{\sigma T^4}{\pi}. \quad (3.28)$$

In Problem 3.14, you will use Eq. (3.27) to express the Stefan–Boltzmann constant, σ , in terms of the fundamental constants c , h , and k . The monochromatic luminosity is related to the **monochromatic flux**, $F_\lambda d\lambda$, by the inverse square law for light, Eq. (3.2):

$$F_\lambda d\lambda = \frac{L_\lambda}{4\pi r^2} d\lambda = \frac{2\pi hc^2/\lambda^5}{e^{hc/\lambda kT} - 1} \left(\frac{R}{r}\right)^2 d\lambda, \quad (3.29)$$

where r is the distance to the model star. Thus $F_\lambda d\lambda$ is the number of joules of starlight energy with wavelengths between λ and $\lambda + d\lambda$ that arrive per second at one square meter of a detector aimed at the model star, assuming that no light has been absorbed or scattered during its journey from the star to the detector. Of course, Earth’s atmosphere absorbs some starlight, but measurements of fluxes and apparent magnitudes can be corrected to account for this absorption; see Section 9.2. The values of these quantities usually quoted for stars are in fact corrected values and would be the results of measurements above Earth’s absorbing atmosphere.

3.6 ■ THE COLOR INDEX

The apparent and absolute magnitudes discussed in Section 3.2, measured over all wavelengths of light emitted by a star, are known as **bolometric magnitudes** and are denoted by m_{bol} and M_{bol} , respectively.²¹ In practice, however, detectors measure the radiant flux of a star only within a certain wavelength region defined by the sensitivity of the detector.

UBV Wavelength Filters

The *color* of a star may be precisely determined by using filters that transmit the star’s light only within certain narrow wavelength bands. In the standard *UBV* system, a star’s apparent magnitude is measured through three filters and is designated by three capital letters:

- U , the star’s *ultraviolet* magnitude, is measured through a filter centered at 365 nm with an effective bandwidth of 68 nm.
- B , the star’s *blue* magnitude, is measured through a filter centered at 440 nm with an effective bandwidth of 98 nm.
- V , the star’s *visual* magnitude, is measured through a filter centered at 550 nm with an effective bandwidth of 89 nm.

Color Indices and the Bolometric Correction

Using Eq. (3.6), a star’s absolute color magnitudes M_U , M_B , and M_V may be determined if its distance d is known.²² A star’s **$U - B$ color index** is the difference between its ultraviolet

²¹A *bolometer* is an instrument that measures the increase in temperature caused by the radiant flux it receives at all wavelengths.

²²Note that although apparent magnitude is not denoted by a subscripted “ m ” in the *UBV* system, the absolute magnitude is denoted by a subscripted “ M .”

and blue magnitudes, and a star's $B - V$ color index is the difference between its blue and visual magnitudes:

$$U - B = M_U - M_B$$

and

$$B - V = M_B - M_V.$$

Stellar magnitudes *decrease* with increasing brightness; consequently, a star with a smaller $B - V$ color index is *bluer* than a star with a larger value of $B - V$. Because a color index is the difference between two magnitudes, Eq. (3.6) shows that it is independent of the star's distance. The difference between a star's bolometric magnitude and its visual magnitude is its **bolometric correction** BC :

$$BC = m_{\text{bol}} - V = M_{\text{bol}} - M_V. \quad (3.30)$$

Example 3.6.1. Sirius, the brightest-appearing star in the sky, has U , B , and V apparent magnitudes of $U = -1.47$, $B = -1.43$, and $V = -1.44$. Thus for Sirius,

$$U - B = -1.47 - (-1.43) = -0.04$$

and

$$B - V = -1.43 - (-1.44) = 0.01.$$

Sirius is brightest at ultraviolet wavelengths, as expected for a star with an effective temperature of $T_e = 9970$ K. For this surface temperature,

$$\lambda_{\text{max}} = \frac{0.0029 \text{ m K}}{9970 \text{ K}} = 291 \text{ nm},$$

which is in the ultraviolet portion of the electromagnetic spectrum. The bolometric correction for Sirius is $BC = -0.09$, so its apparent bolometric magnitude is

$$m_{\text{bol}} = V + BC = -1.44 + (-0.09) = -1.53.$$

The relation between apparent magnitude and radiant flux, Eq. (3.4), can be used to derive expressions for the ultraviolet, blue, and visual magnitudes measured (above Earth's atmosphere) for a star. A *sensitivity function* $\mathcal{S}(\lambda)$ is used to describe the fraction of the star's flux that is detected at wavelength λ . $\mathcal{S}$ depends on the reflectivity of the telescope mirrors, the bandwidth of the U , B , and V filters, and the response of the photometer. Thus, for example, a star's ultraviolet magnitude U is given by

$$U = -2.5 \log_{10} \left(\int_0^\infty F_\lambda \mathcal{S}_U d\lambda \right) + C_U, \quad (3.31)$$

where C_U is a constant. Similar expressions are used for a star's apparent magnitude within other wavelength bands. The constants C in the equations for U , B , and V differ for each

of these wavelength regions and are chosen so that the star Vega (α Lyrae) has a magnitude of *zero* as seen through each filter.²³ This is a completely arbitrary choice and does *not* imply that Vega would appear equally bright when viewed through the U , B , and V filters. However, the resulting values for the visual magnitudes of stars are about the same as those recorded by Hipparchus two thousand years ago.²⁴

A *different* method is used to determine the constant C_{bol} in the expression for the bolometric magnitude, measured over all wavelengths of light emitted by a star. For a *perfect* bolometer, capable of detecting 100 percent of the light arriving from a star, we set $\mathcal{S}(\lambda) \equiv 1$:

$$m_{\text{bol}} = -2.5 \log_{10} \left(\int_0^\infty F_\lambda d\lambda \right) + C_{\text{bol}}. \quad (3.32)$$

The value for C_{bol} originated in the wish of astronomers that the value of the bolometric correction

$$BC = m_{\text{bol}} - V$$

be negative for all stars (since a star's radiant flux over all wavelengths is greater than its flux in any specified wavelength band) while still being as close to zero as possible. After a value of C_{bol} was agreed upon, it was discovered that some supergiant stars have *positive* bolometric corrections. Nevertheless, astronomers have chosen to continue using this unphysical method of measuring magnitudes.²⁵ It is left as an exercise for you to evaluate the constant C_{bol} by using the value of m_{bol} assigned to the Sun: $m_{\text{Sun}} = -26.83$.

The color indices $U - B$ and $B - V$ are immediately seen to be

$$U - B = -2.5 \log_{10} \left(\frac{\int F_\lambda \mathcal{S}_U d\lambda}{\int F_\lambda \mathcal{S}_B d\lambda} \right) + C_{U-B}, \quad (3.33)$$

where $C_{U-B} \equiv C_U - C_B$. A similar relation holds for $B - V$. From Eq. (3.29), note that although the apparent magnitudes depend on the radius R of the model star and its distance r , the color indices do not, because the factor of $(R/r)^2$ cancels in Eq. (3.33). Thus the color index is a measure solely of the temperature of a model blackbody star.

Example 3.6.2. A certain hot star has a surface temperature of 42,000 K and color indices $U - B = -1.19$ and $B - V = -0.33$. The large negative value of $U - B$ indicates that this star appears brightest at ultraviolet wavelengths, as can be confirmed with Wien's displacement law, Eq. (3.19). The spectrum of a 42,000-K blackbody peaks at

$$\lambda_{\text{max}} = \frac{0.0029 \text{ m K}}{42,000 \text{ K}} = 69 \text{ nm},$$

continued

²³Actually, the average magnitude of several stars is used for this calibration.

²⁴See Chapter 1 of Böhm-Vitense (1989b) for a further discussion of the vagaries of the magnitude system used by astronomers.

²⁵Some authors, such as Böhm-Vitense (1989a, 1989b), prefer to define the bolometric correction as $BC = V - m_{\text{bol}}$, so their values of BC are usually positive.

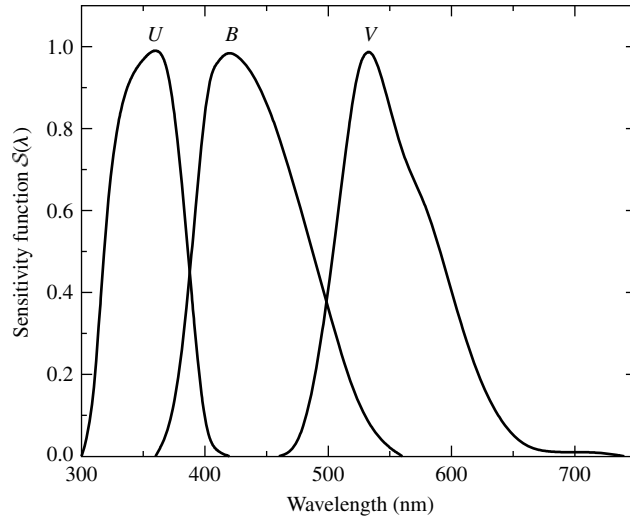


FIGURE 3.10 Sensitivity functions $S(\lambda)$ for U , B , and V filters. (Data from Johnson, *Ap. J.*, 141, 923, 1965.)

in the ultraviolet region of the electromagnetic spectrum. This wavelength is much shorter than the wavelengths transmitted by the U , B , and V filters (see Fig. 3.10), so we will be dealing with the smoothly declining long-wavelength “tail” of the Planck function $B_\lambda(T)$.

We can use the values of the color indices to estimate the constant C_{U-B} in Eq. (3.33), and C_{B-V} in a similar equation for the color index $B - V$. In this estimate, we will use a step function to represent the sensitivity function: $S(\lambda) = 1$ inside the filter’s bandwidth, and $S(\lambda) = 0$ otherwise. The integrals in Eq. (3.33) may then be approximated by the value of the Planck function B_λ at the center of the filter bandwidth, multiplied by that bandwidth. Thus, for the wavelengths and bandwidths $\Delta\lambda$ listed on page 75,

$$U - B = -2.5 \log_{10} \left(\frac{B_{365} \Delta\lambda_U}{B_{440} \Delta\lambda_B} \right) + C_{U-B}$$

$$-1.19 = -0.32 + C_{U-B}$$

$$C_{U-B} = -0.87,$$

and

$$B - V = -2.5 \log_{10} \left(\frac{B_{440} \Delta\lambda_B}{B_{550} \Delta\lambda_V} \right) + C_{B-V}$$

$$-0.33 = -0.98 + C_{B-V}$$

$$C_{B-V} = 0.65.$$

It is left as an exercise for you to use these values of C_{U-B} and C_{B-V} to estimate the color indices for a model blackbody Sun with a surface temperature of 5777 K. Although

the resulting value of $B - V = +0.57$ is in fair agreement with the measured value of $B - V = +0.650$ for the Sun, the estimate of $U - B = -0.22$ is quite different from the measured value of $U - B = +0.195$. The reason for this large discrepancy at ultraviolet wavelengths will be discussed in Example 9.2.4.

The Color–Color Diagram

Figure 3.11 is a **color–color diagram** showing the relation between the $U - B$ and $B - V$ color indices for main-sequence stars.²⁶ Astronomers face the difficult task of connecting a star’s position on a color–color diagram with the physical properties of the star itself. If stars actually behaved as blackbodies, the color–color diagram would be the straight dashed line shown in Fig. 3.11. However, stars are not true blackbodies. As will be discussed in detail in Chapter 9, some light is absorbed as it travels through a star’s atmosphere, and the amount of light absorbed depends on both the wavelength of the light and the temperature of the star. Other factors also play a role, causing the color indices of main-sequence and supergiant stars of the same temperature to be slightly different. The color–color diagram in Fig. 3.11 shows that the agreement between actual stars and model blackbody stars is best for very hot stars.

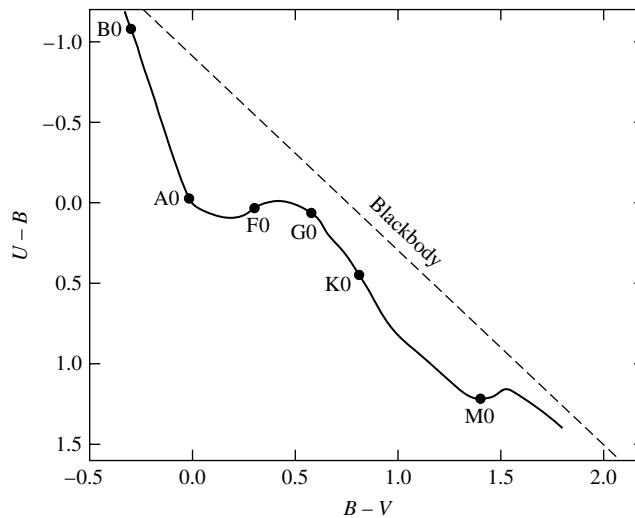


FIGURE 3.11 Color–color diagram for main-sequence stars. The dashed line is for a blackbody. (The data are taken from Appendix G.)

²⁶As will be discussed in Section 10.6, main-sequence stars are powered by the nuclear fusion of hydrogen nuclei in their centers. Approximately 80% to 90% of all stars are main-sequence stars. The letter labels in Fig. 3.11 are *spectral types*; see Section 8.1.

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PROBLEMS

- 3.1** In 1672, an international effort was made to measure the parallax angle of Mars at the time of opposition, when it was closest to Earth; see Fig. 1.6.
- (a) Consider two observers who are separated by a baseline equal to Earth’s diameter. If the difference in their measurements of Mars’s angular position is $33.6''$, what is the distance between Earth and Mars at the time of opposition? Express your answer both in units of m and in AU.
- (b) If the distance to Mars is to be measured to within 10%, how closely must the clocks used by the two observers be synchronized? *Hint:* Ignore the rotation of Earth. The average orbital velocities of Earth and Mars are 29.79 km s^{-1} and 24.13 km s^{-1} , respectively.